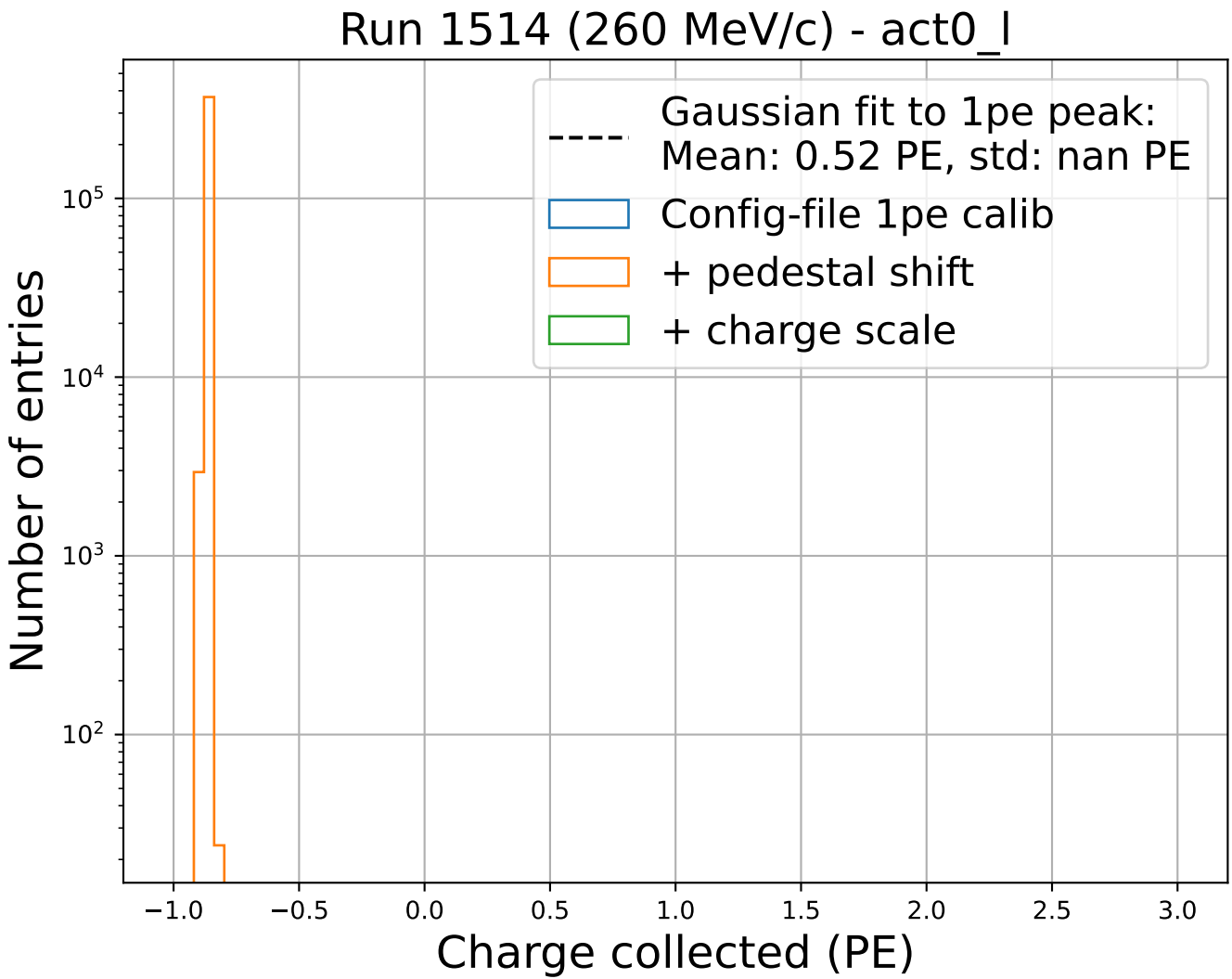
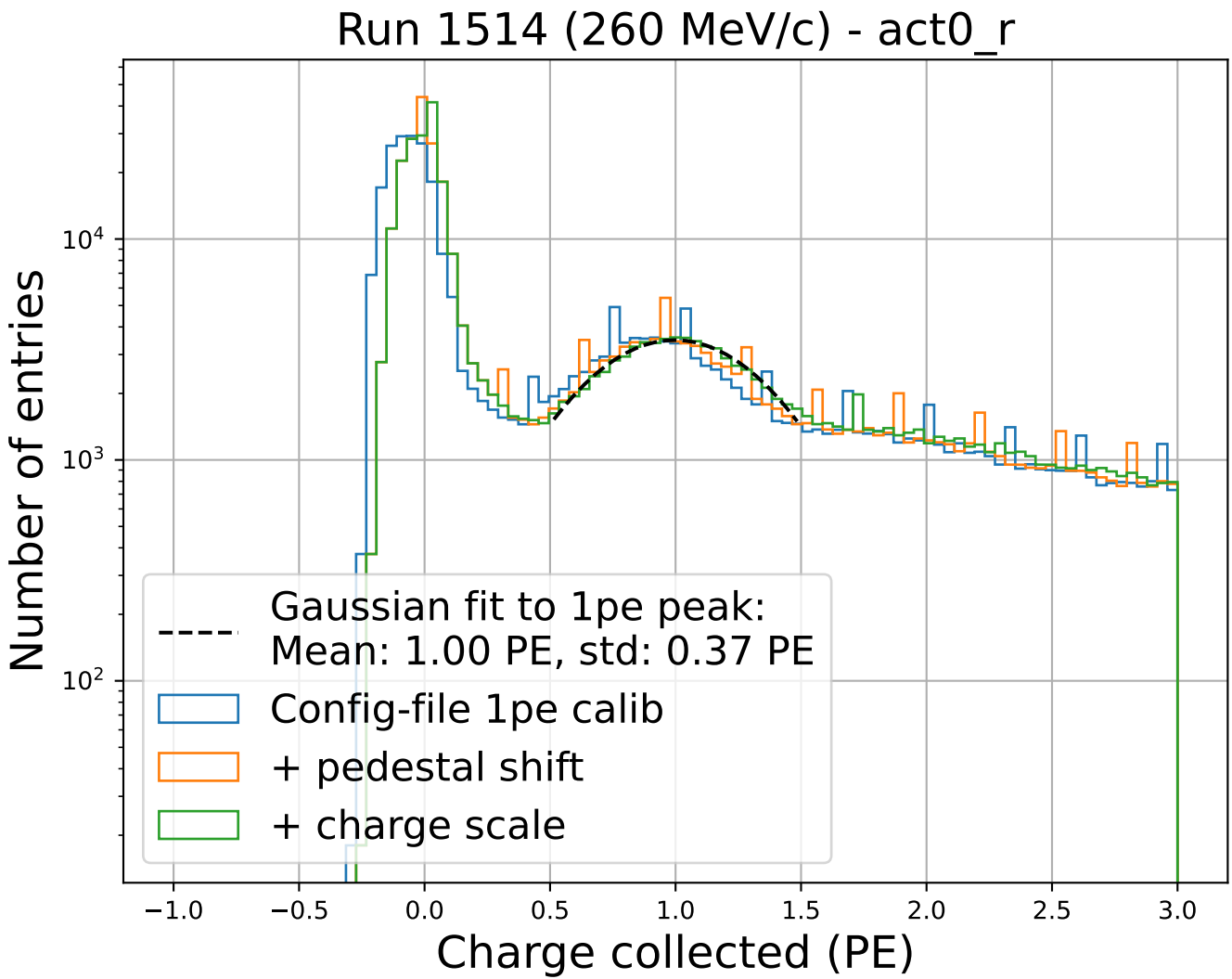
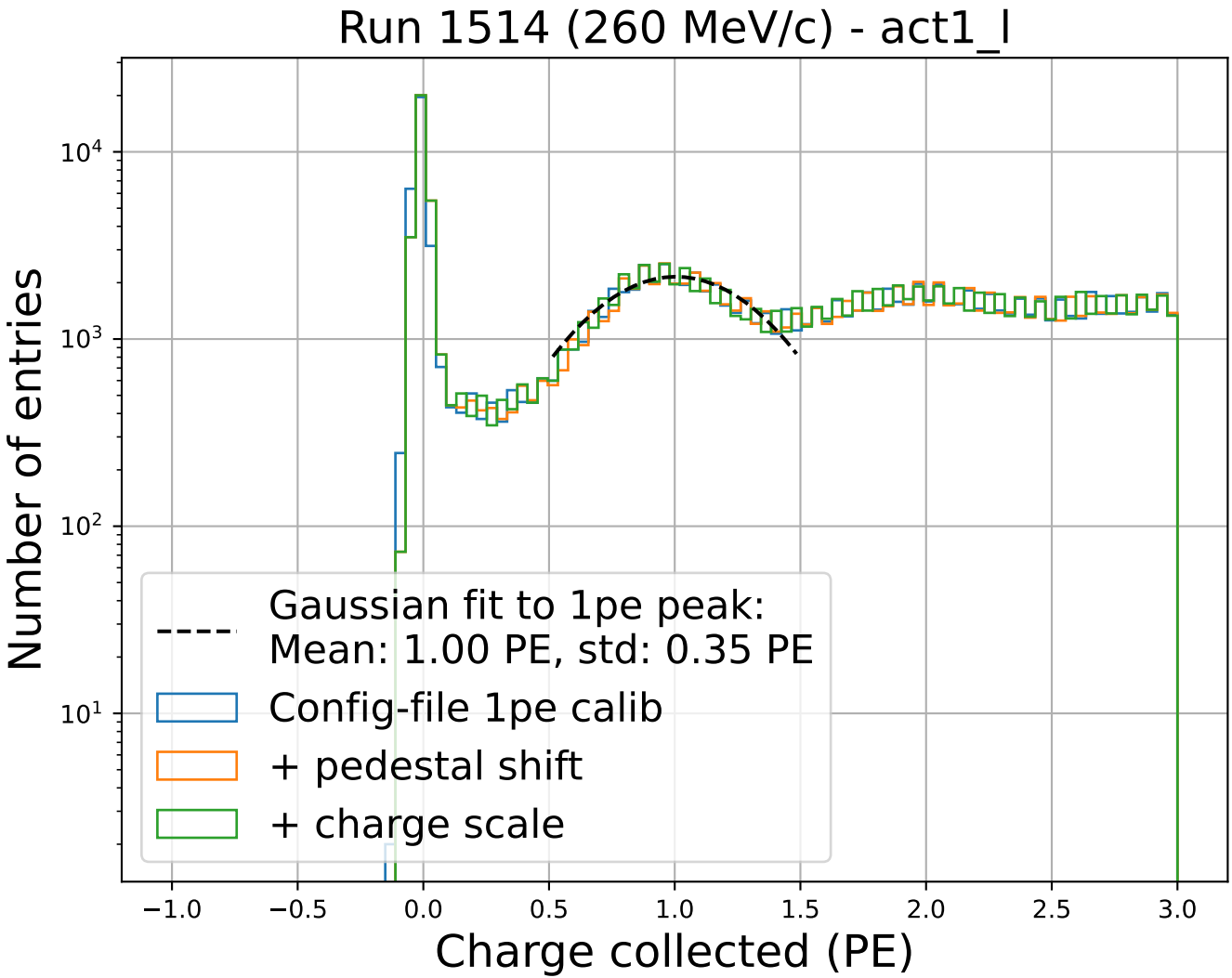
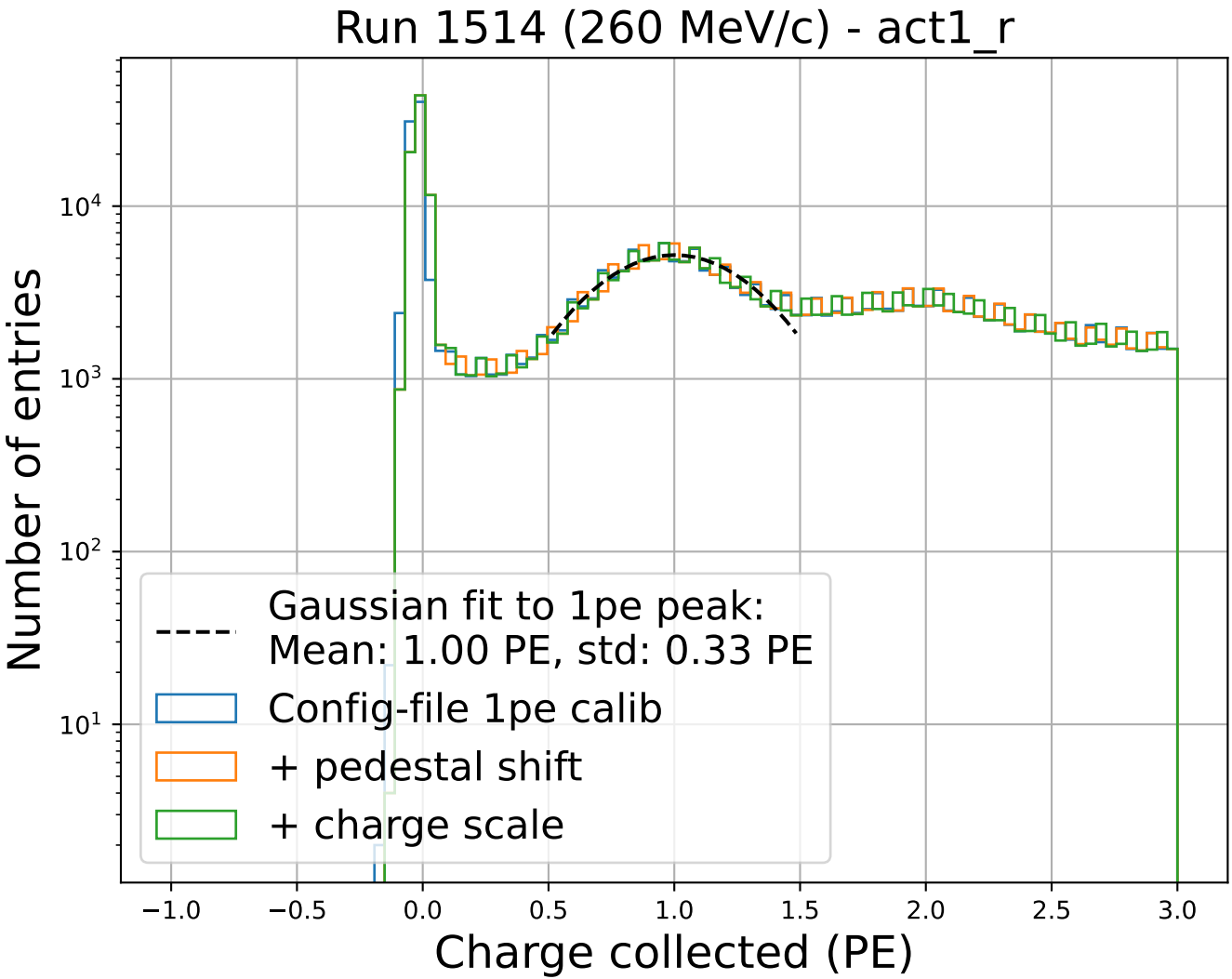
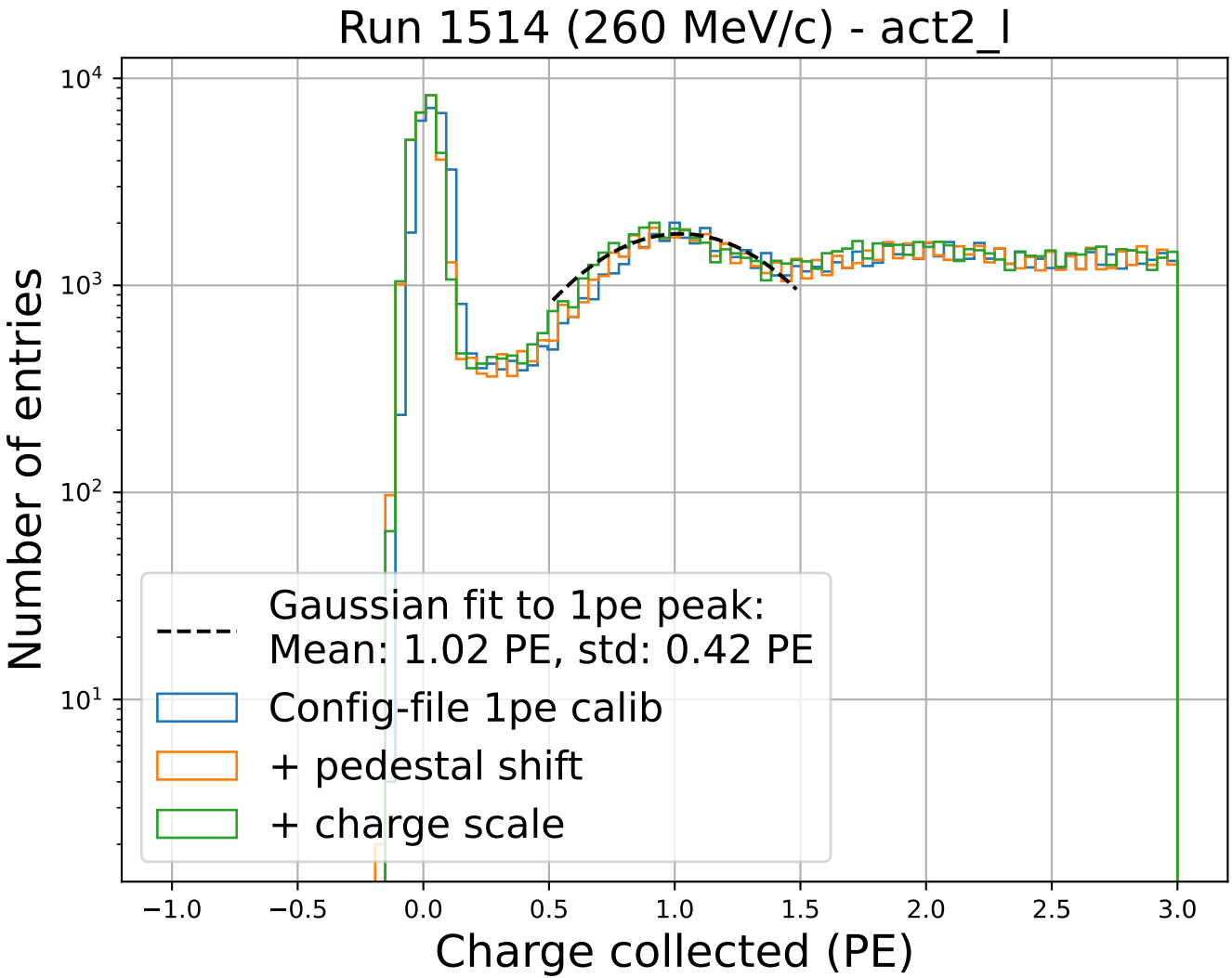
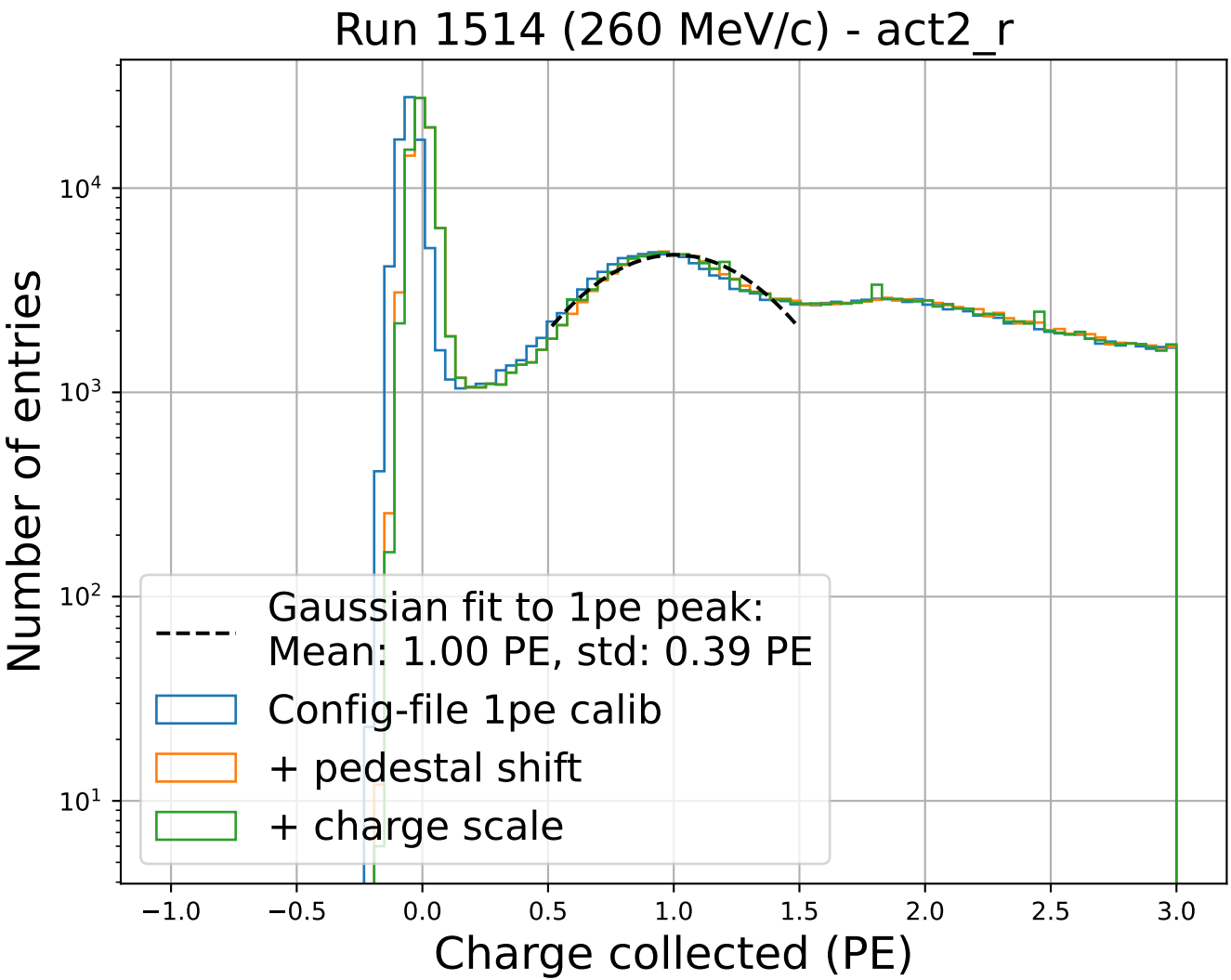

Run 1514 (260 MeV/c) - act0_l
Gaussian fit to 1pe peak:
Mean: 0.52 PE, std: nan PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
Number of entries
10^5
10^4
10^3
10^2
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Charge collected (PE)

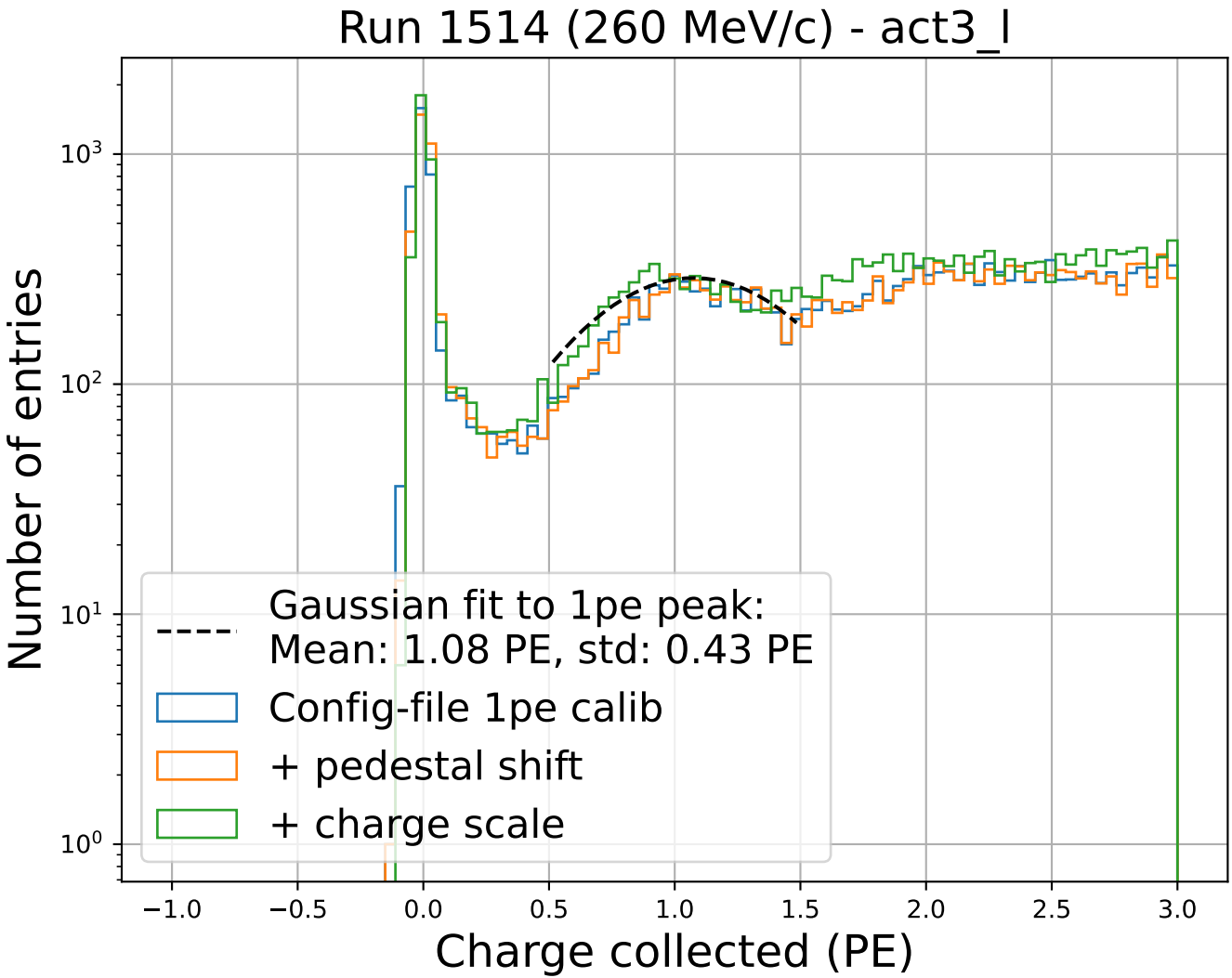

Run 1514 (260 MeV/c) - act0_r
Number of entries
Charge collected (PE)
10^4
10^3
10^2
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.37 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale

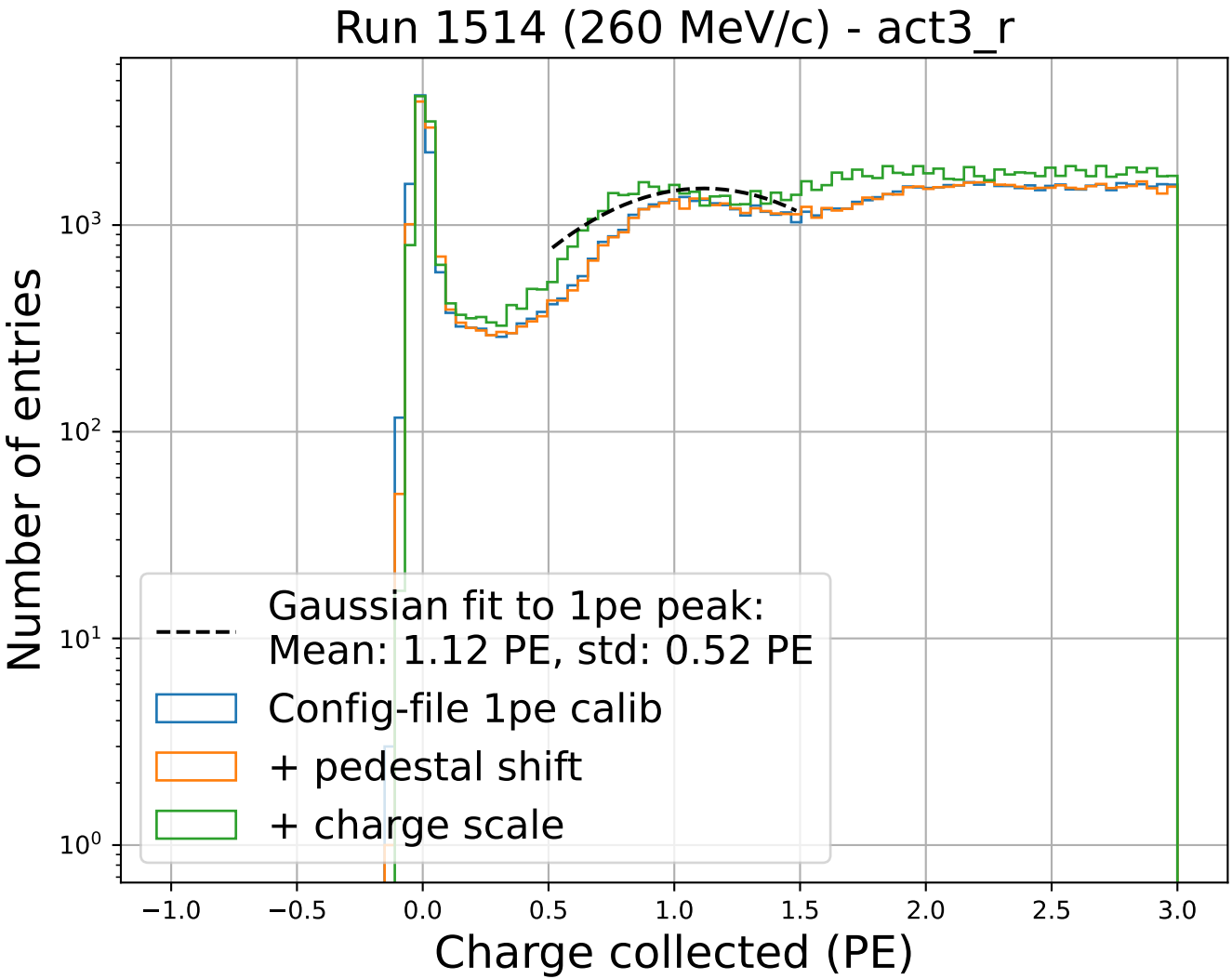

Run 1514 (260 MeV/c) - act1_l
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.35 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale

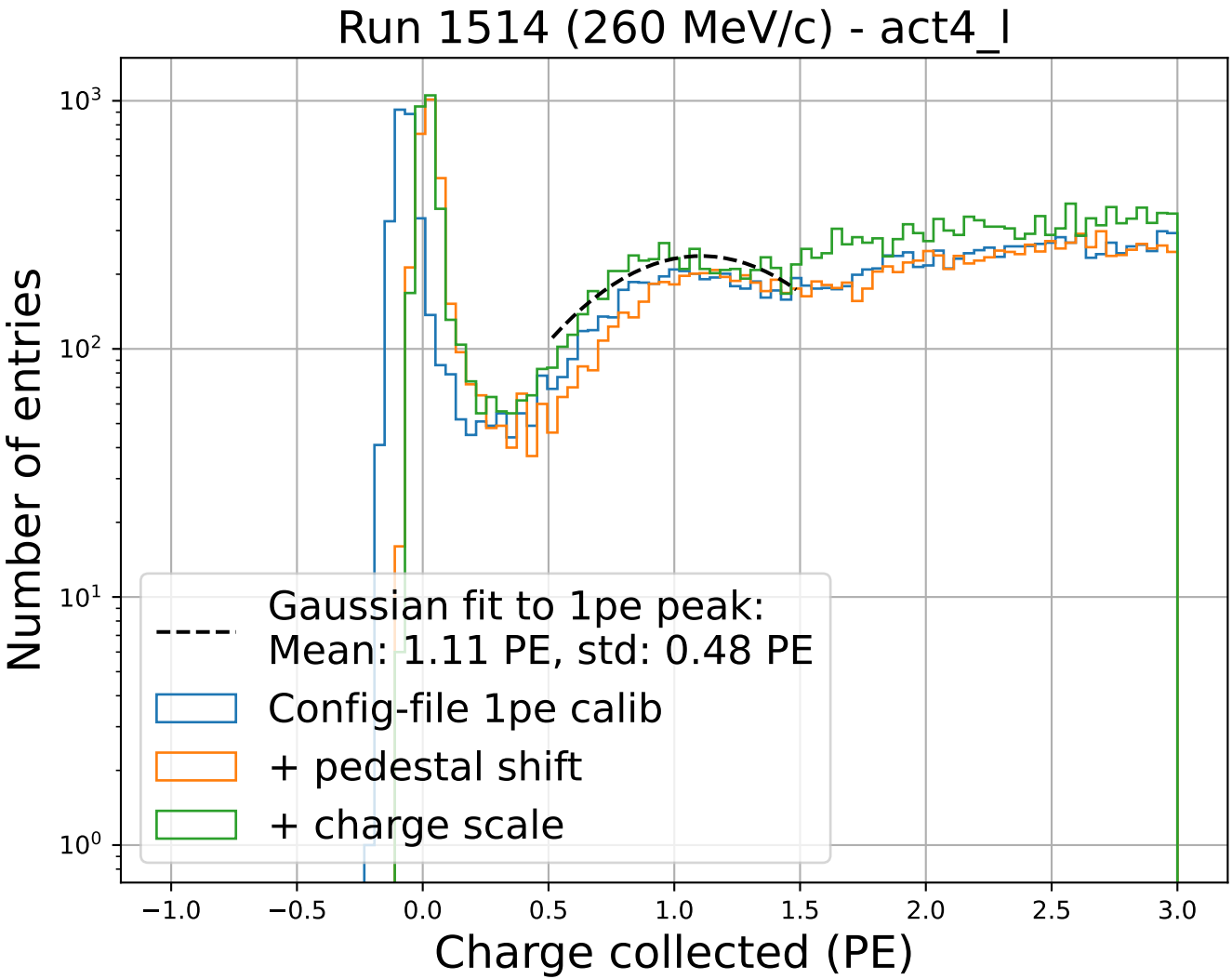

Run 1514 (260 MeV/c) - act1_r
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.33 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
10^4
10^3
10^2
10^1
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

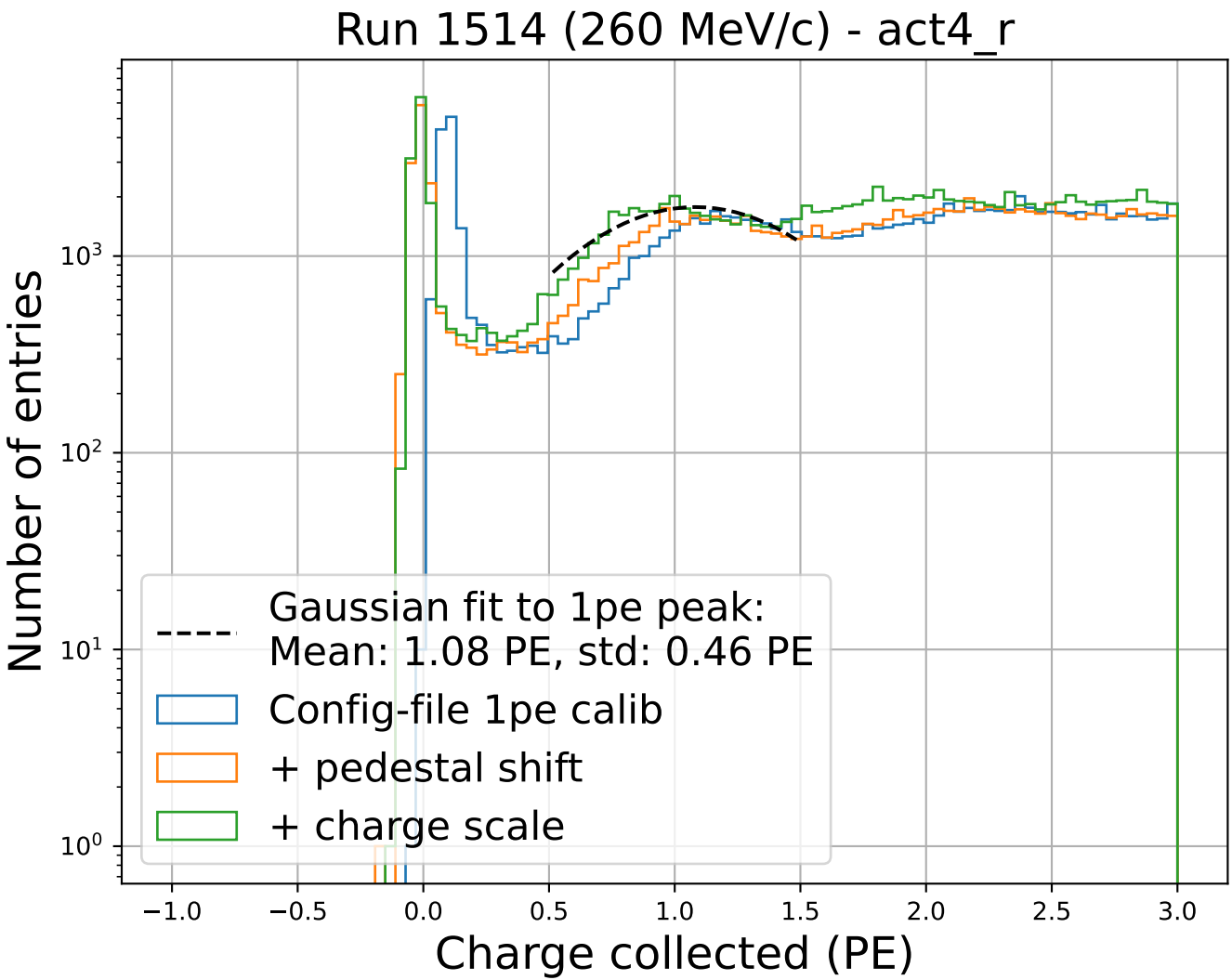

Run 1514 (260 MeV/c) - act2_l
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.02 PE, std: 0.42 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale

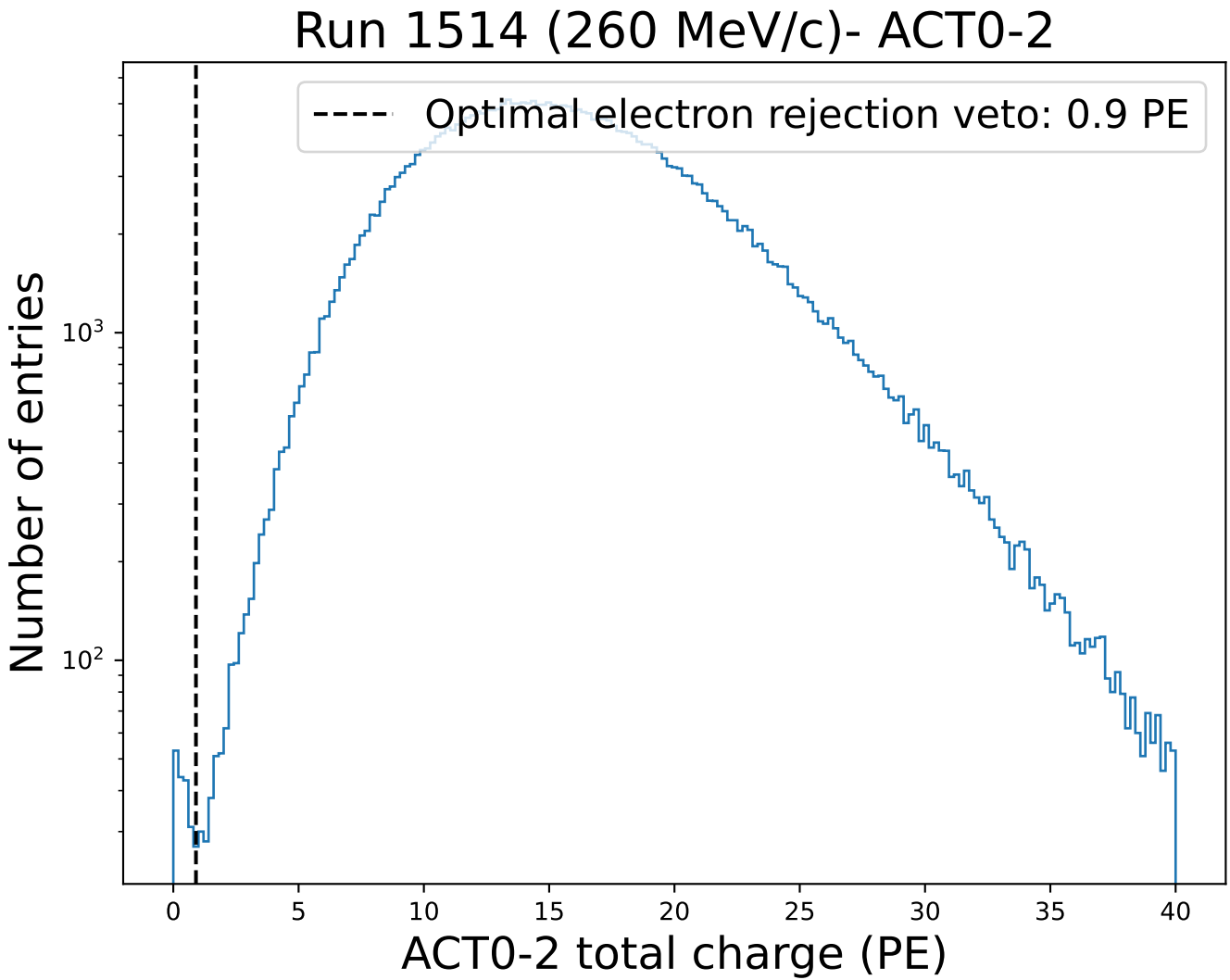

Run 1514 (260 MeV/c) - act2_r
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.00 PE, std: 0.39 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
10^4
10^3
10^2
10^1
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

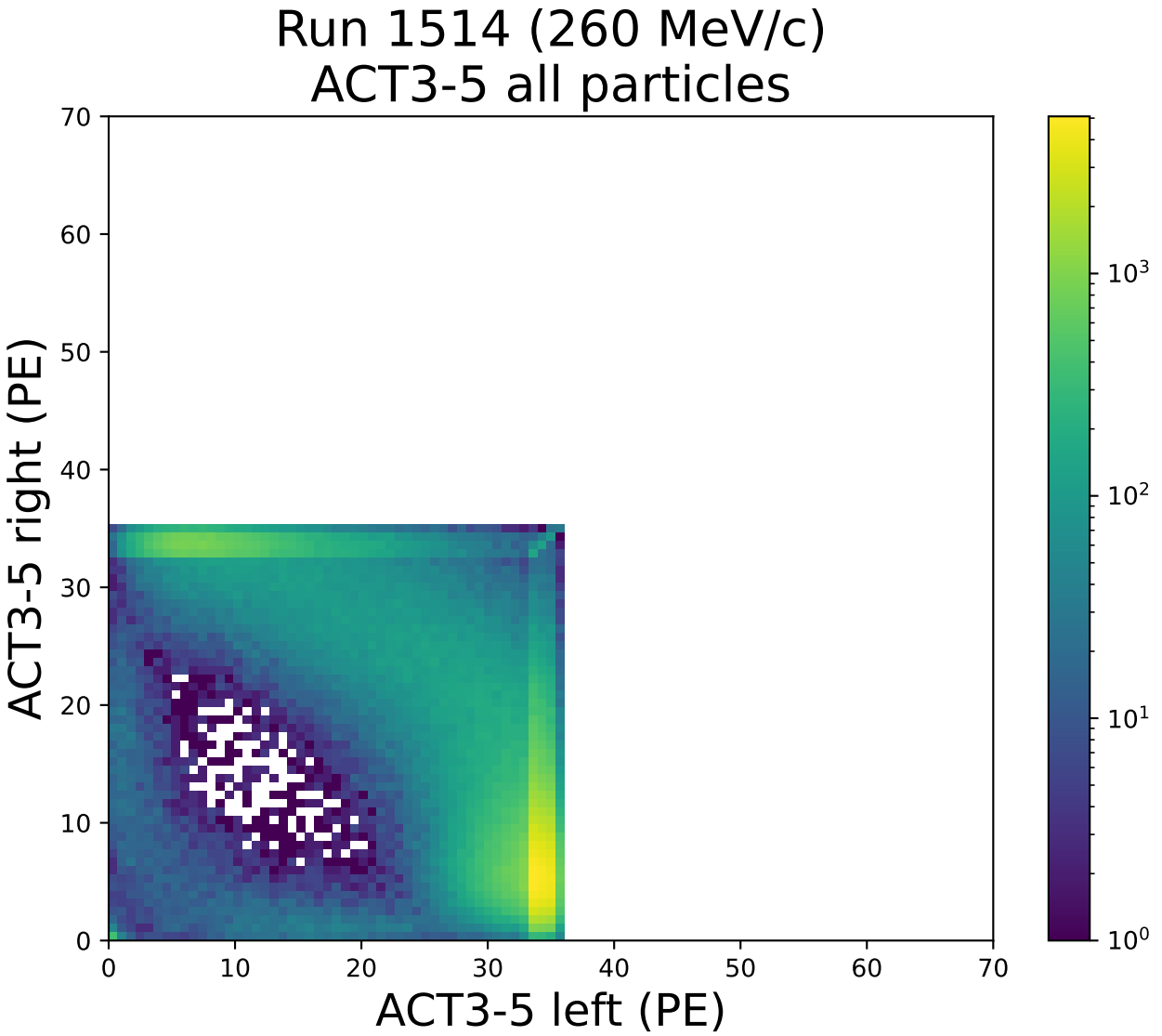

Run 1514 (260 MeV/c) - act3_l
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.08 PE, std: 0.43 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
10^3
10^2
10^1
10^0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

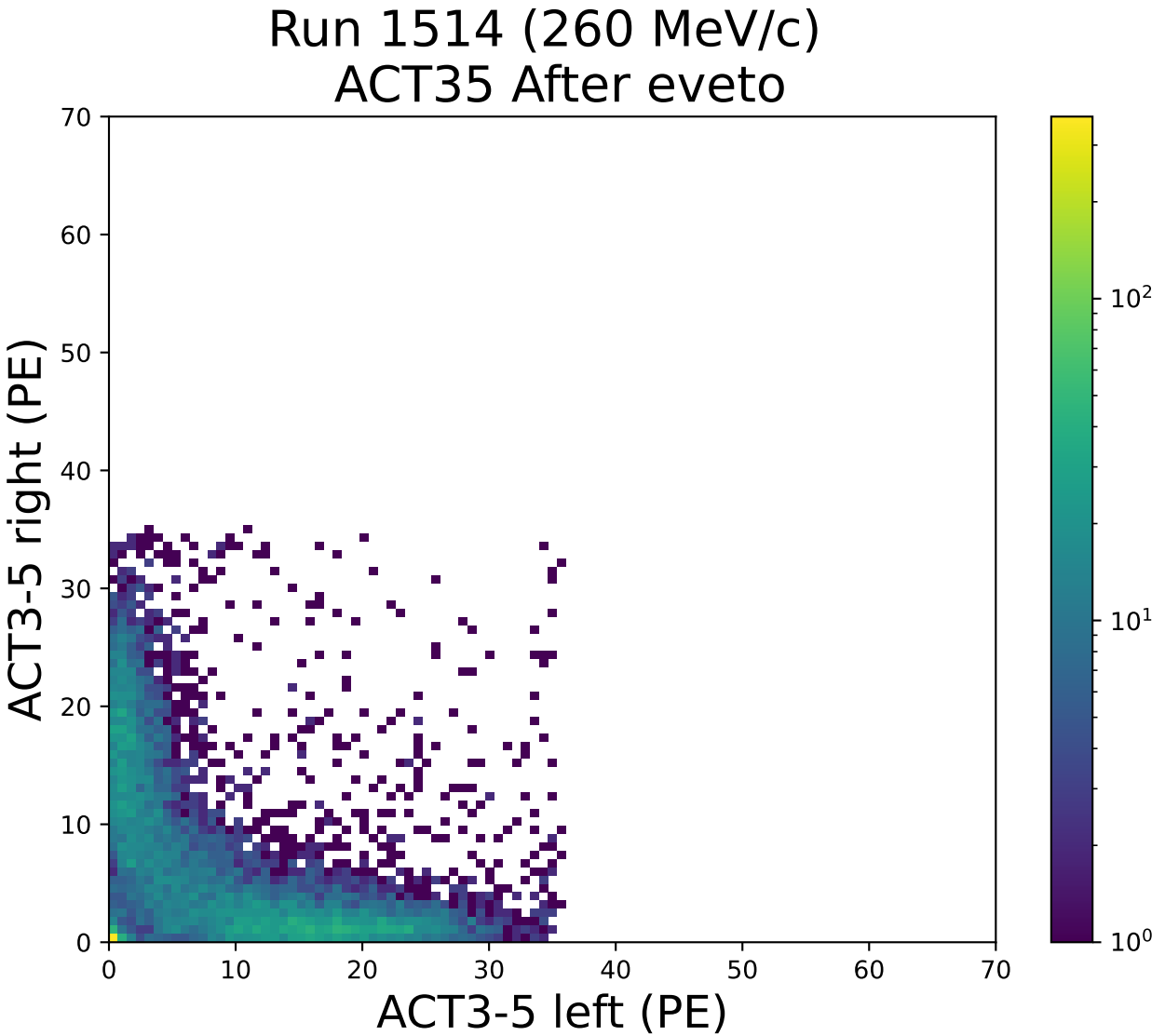

Run 1514 (260 MeV/c) - act3_r
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.12 PE, std: 0.52 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
10^0
10^1
10^2
10^3
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

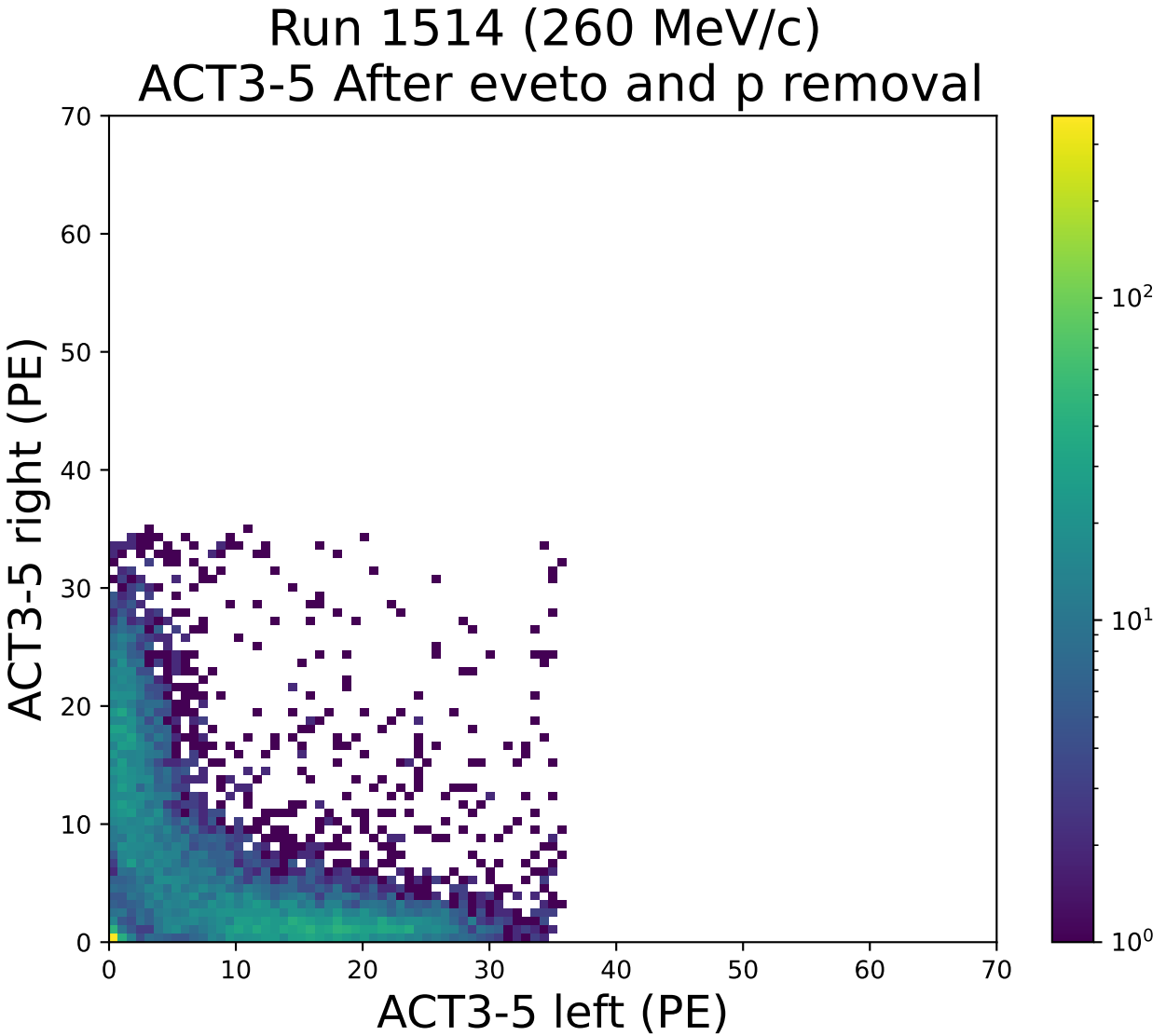

Run 1514 (260 MeV/c) - act4_l
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.11 PE, std: 0.48 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale

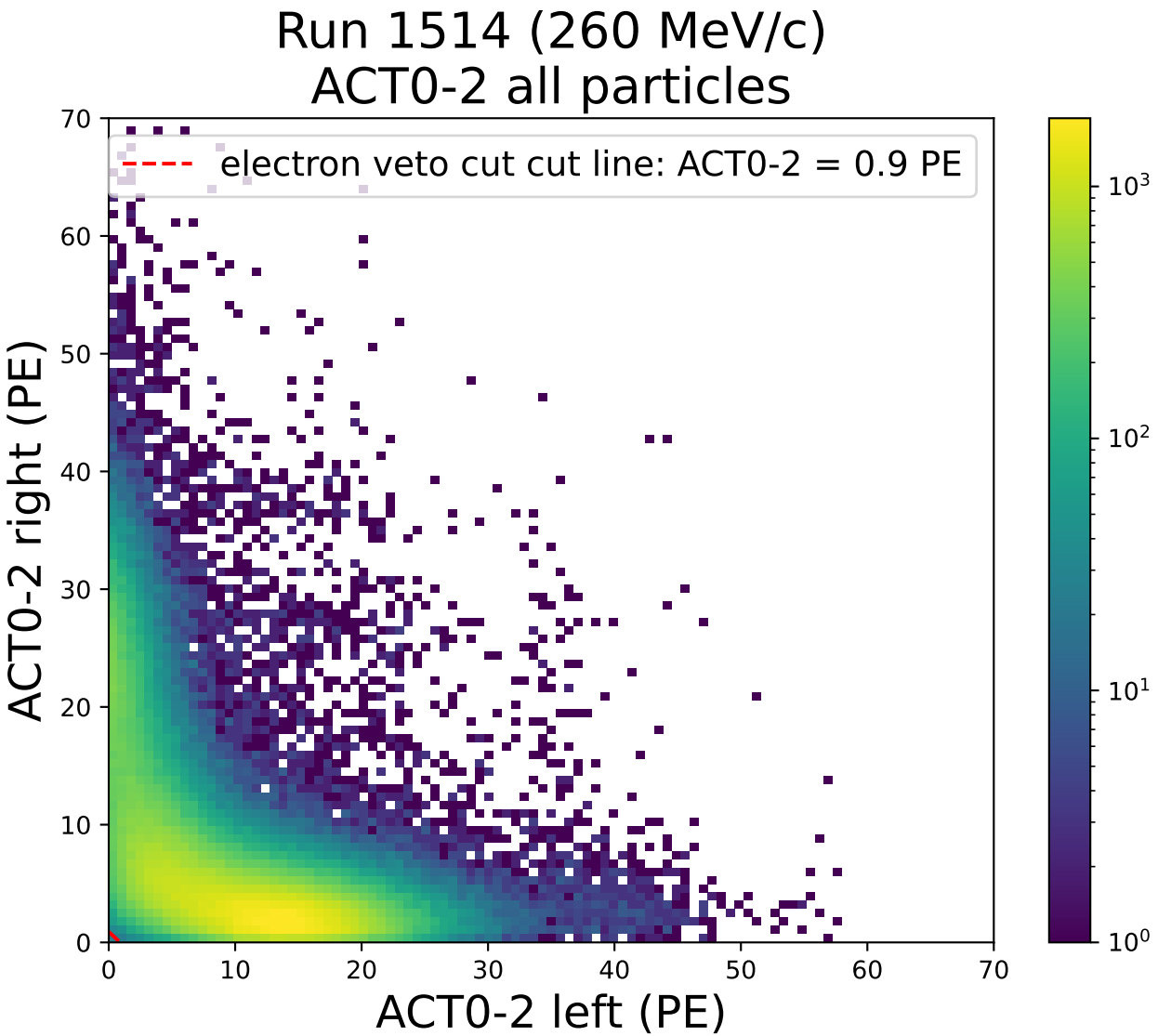

Run 1514 (260 MeV/c) - act4_r
Number of entries
Charge collected (PE)
Gaussian fit to 1pe peak:
Mean: 1.08 PE, std: 0.46 PE
Config-file 1pe calib
+ pedestal shift
+ charge scale
10^0
10^1
10^2
10^3
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

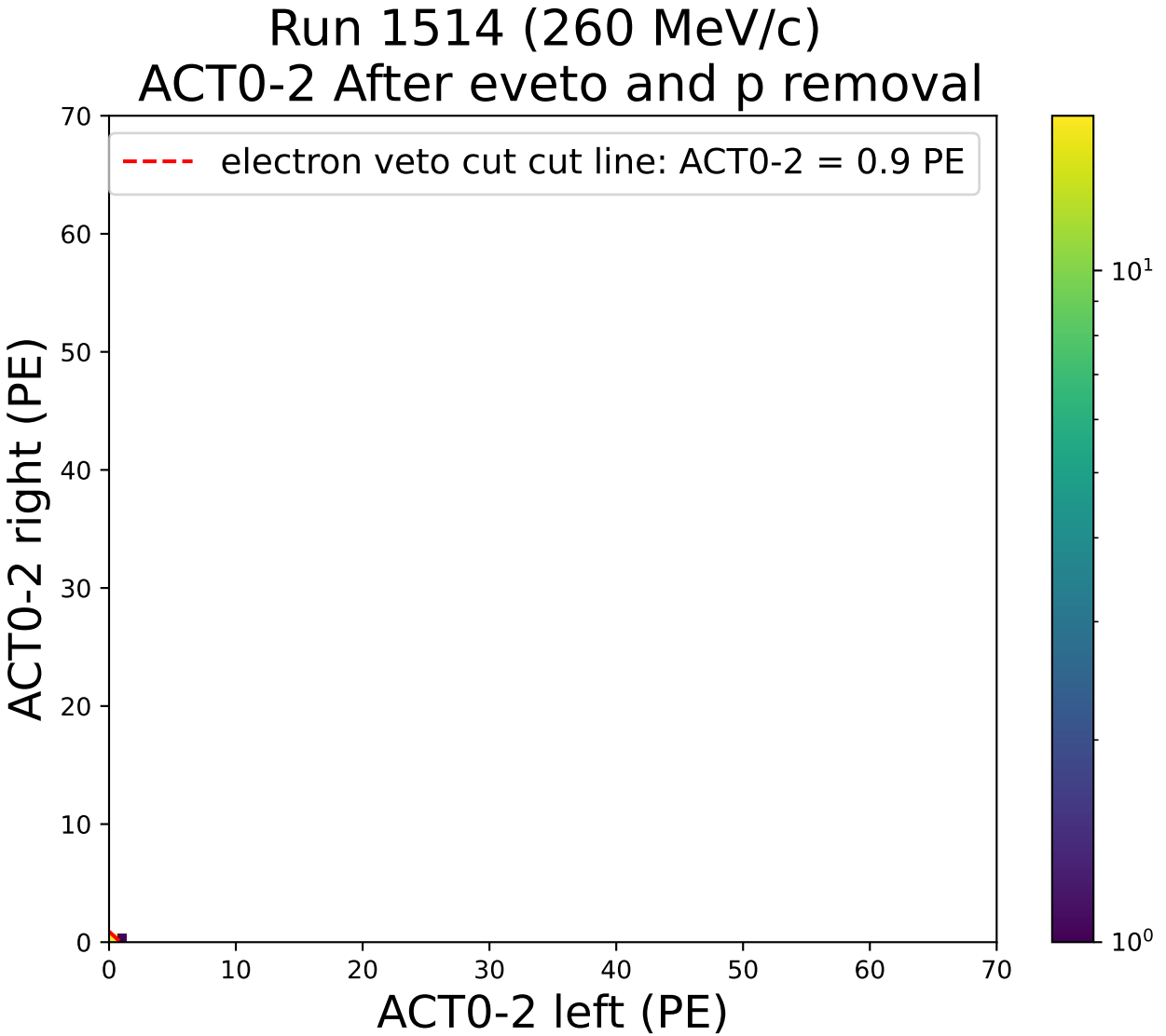

Run 1514 (260 MeV/c)- ACT0-2
Optimal electron rejection veto: 0.9 PE
Number of entries
10^3
10^2
0
5
10
15
20
25
30
35
40
ACT0-2 total charge (PE)

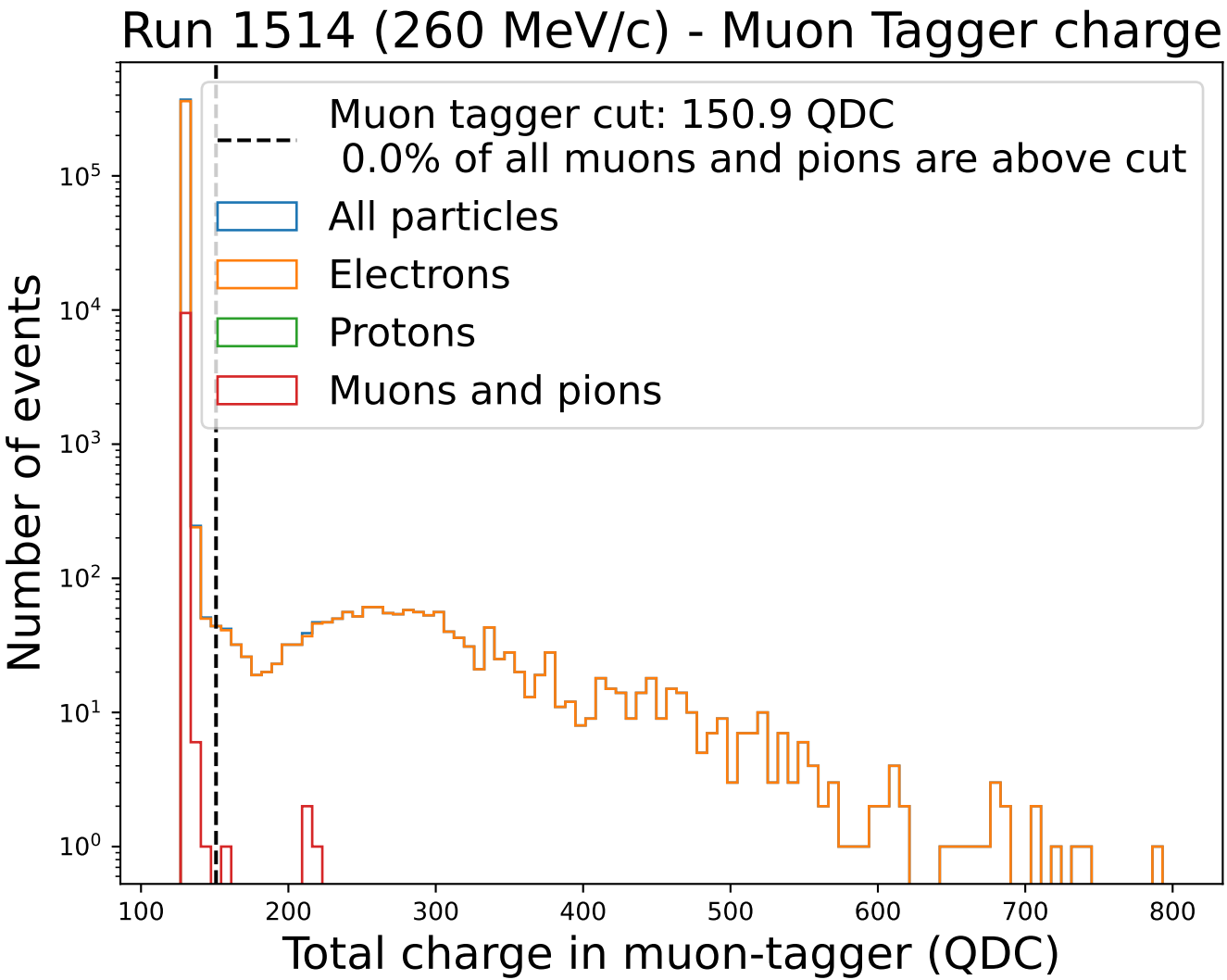

Run 1514 (260 MeV/c)
ACT3-5 all particles
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3

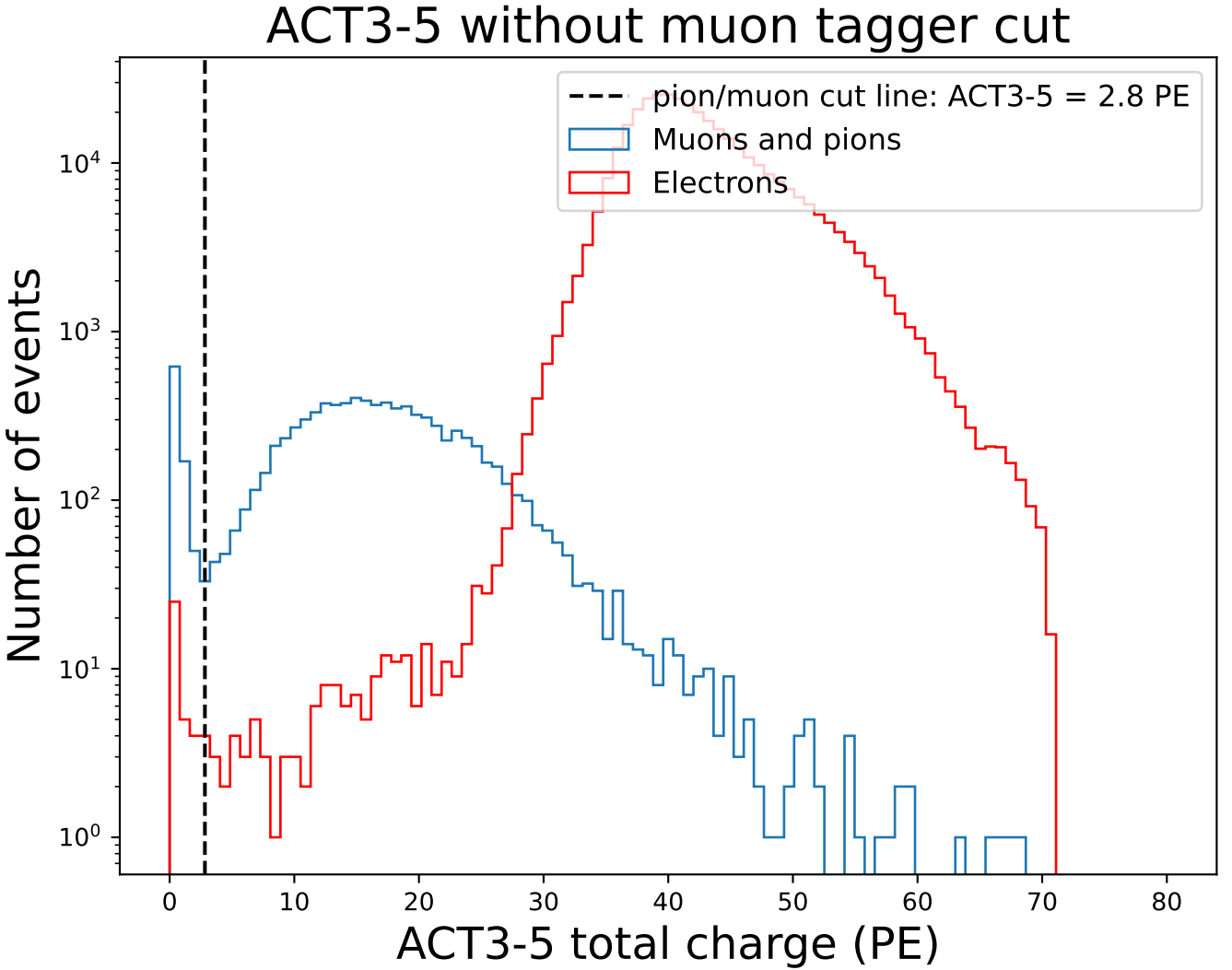

Run 1514 (260 MeV/c)
ACT35 After eveto
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^2
10^1
10^0

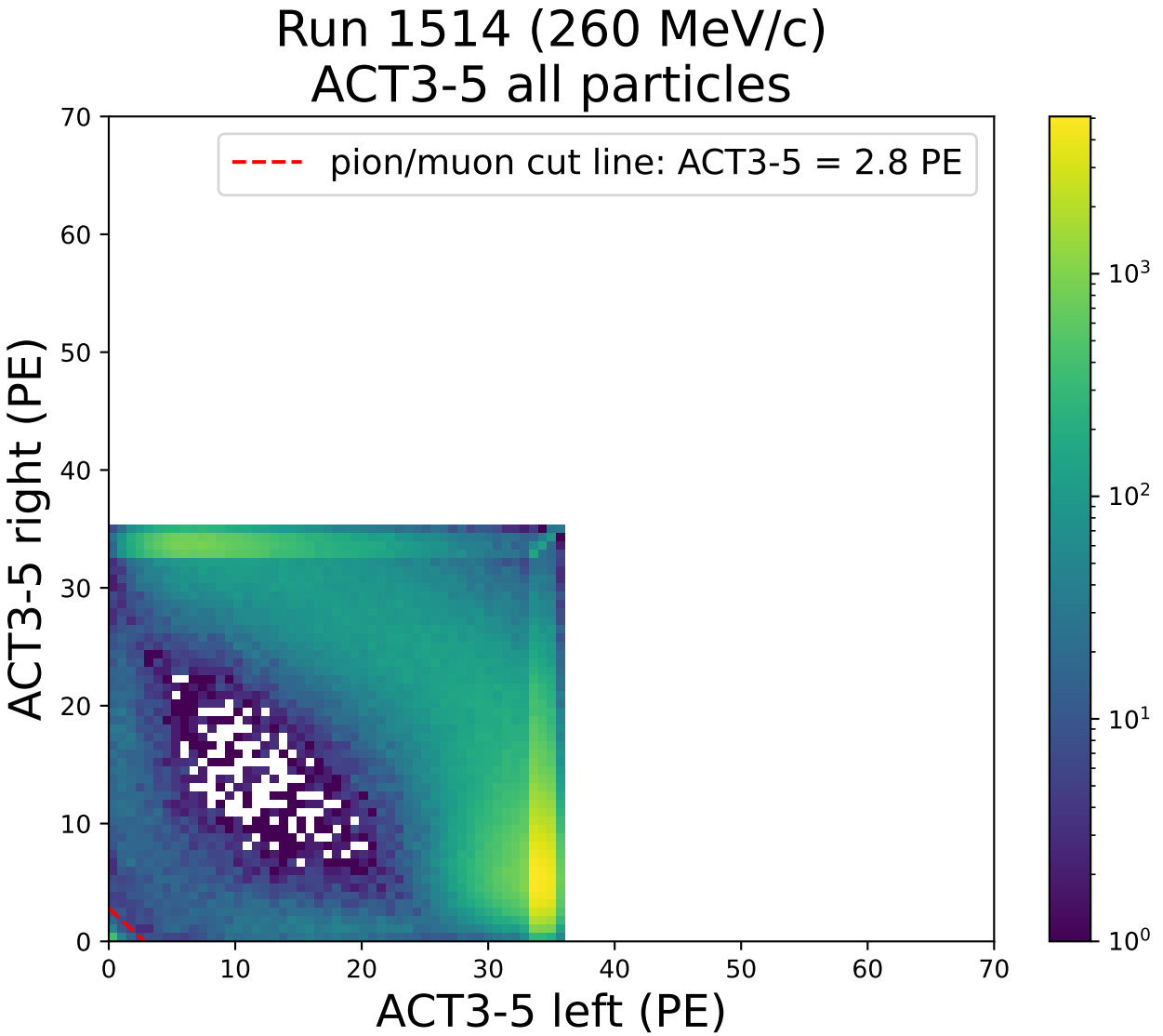

Run 1514 (260 MeV/c)
ACT3-5 After eveto and p removal
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^2
10^1
10^0

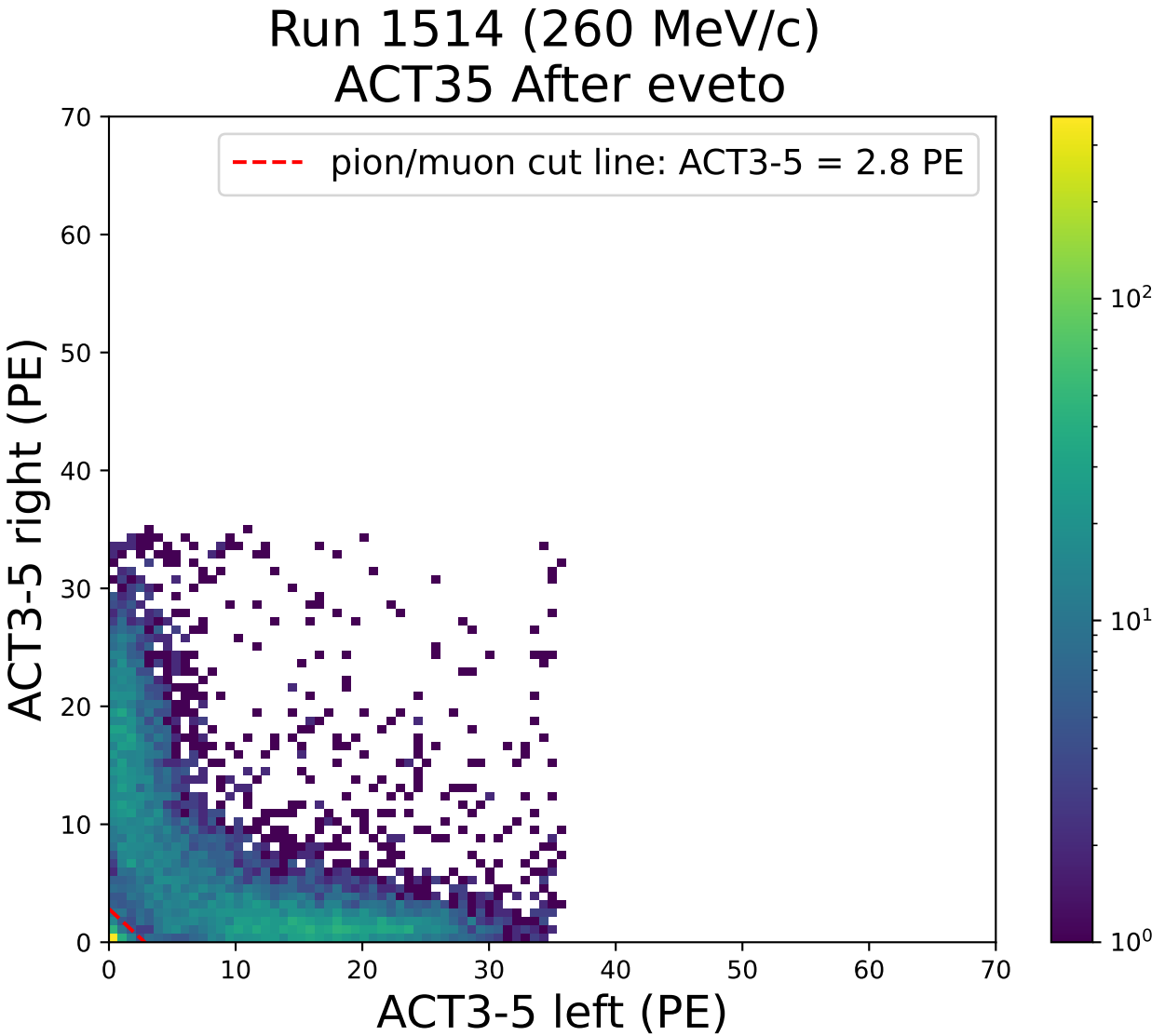

Run 1514 (260 MeV/c)
ACT0-2 all particles
electron veto cut cut line: ACT0-2 = 0.9 PE
ACT0-2 right (PE)
ACT0-2 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3

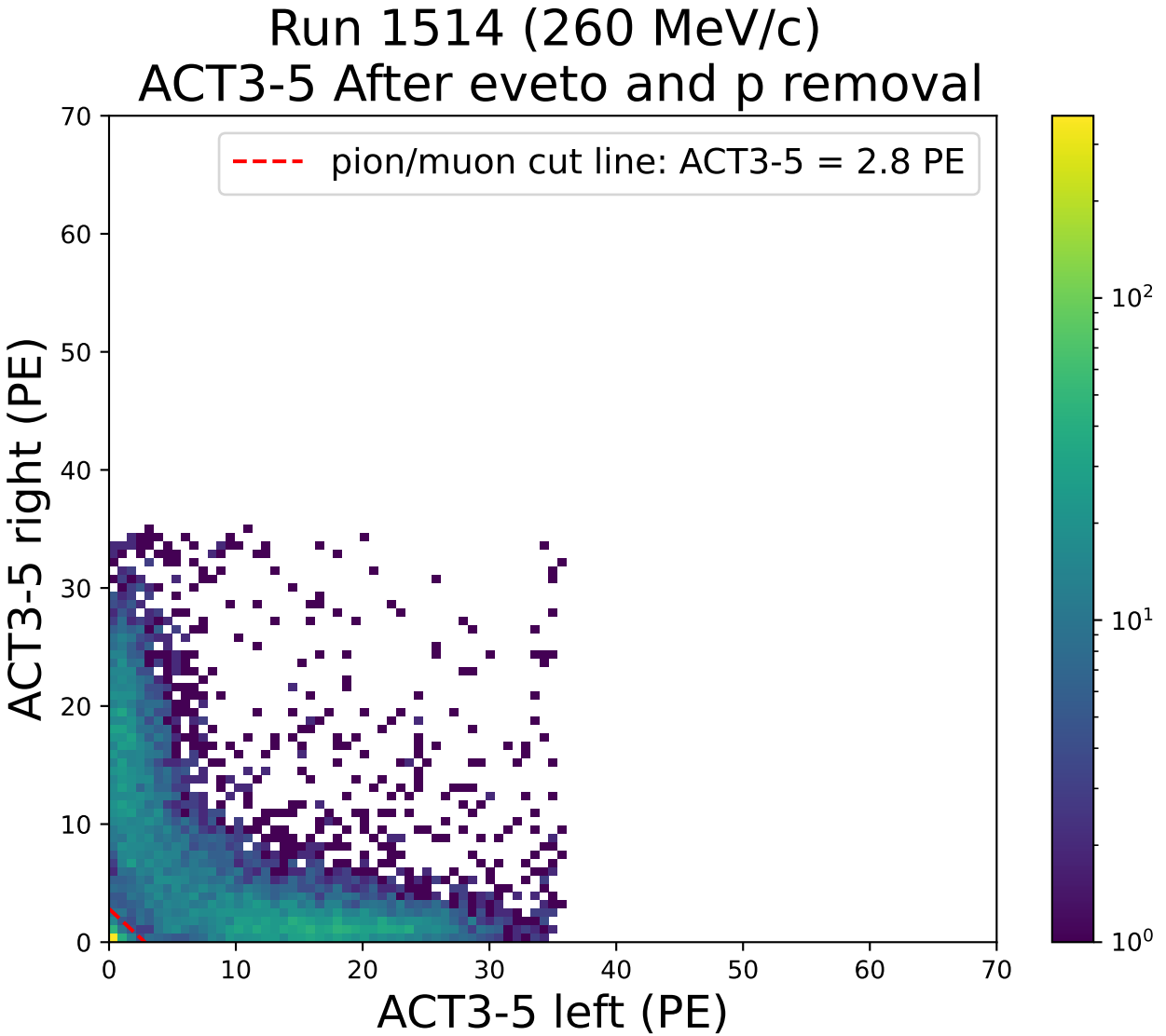

Run 1514 (260 MeV/c)
ACT0-2 After eveto and p removal
electron veto cut cut line: ACT0-2 = 0.9 PE
ACT0-2 right (PE)
ACT0-2 left (PE)
0
10
20
30
40
50
60
70
10^1
10^0

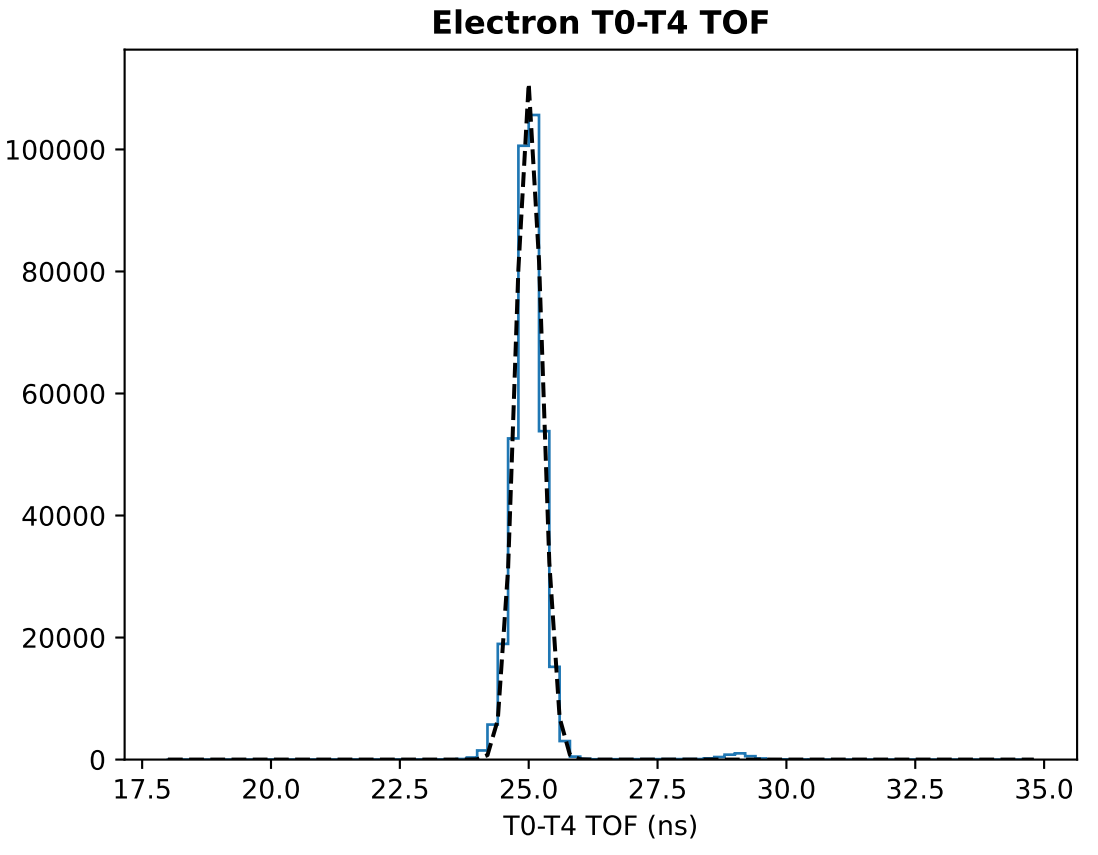

Run 1514 (260 MeV/c) - Muon Tagger charge
Muon tagger cut: 150.9 QDC
0.0% of all muons and pions are above cut
All particles
Electrons
Protons
Muons and pions
Number of events
10^5
10^4
10^3
10^2
10^1
10^0
100
200
300
400
500
600
700
800
Total charge in muon-tagger (QDC)

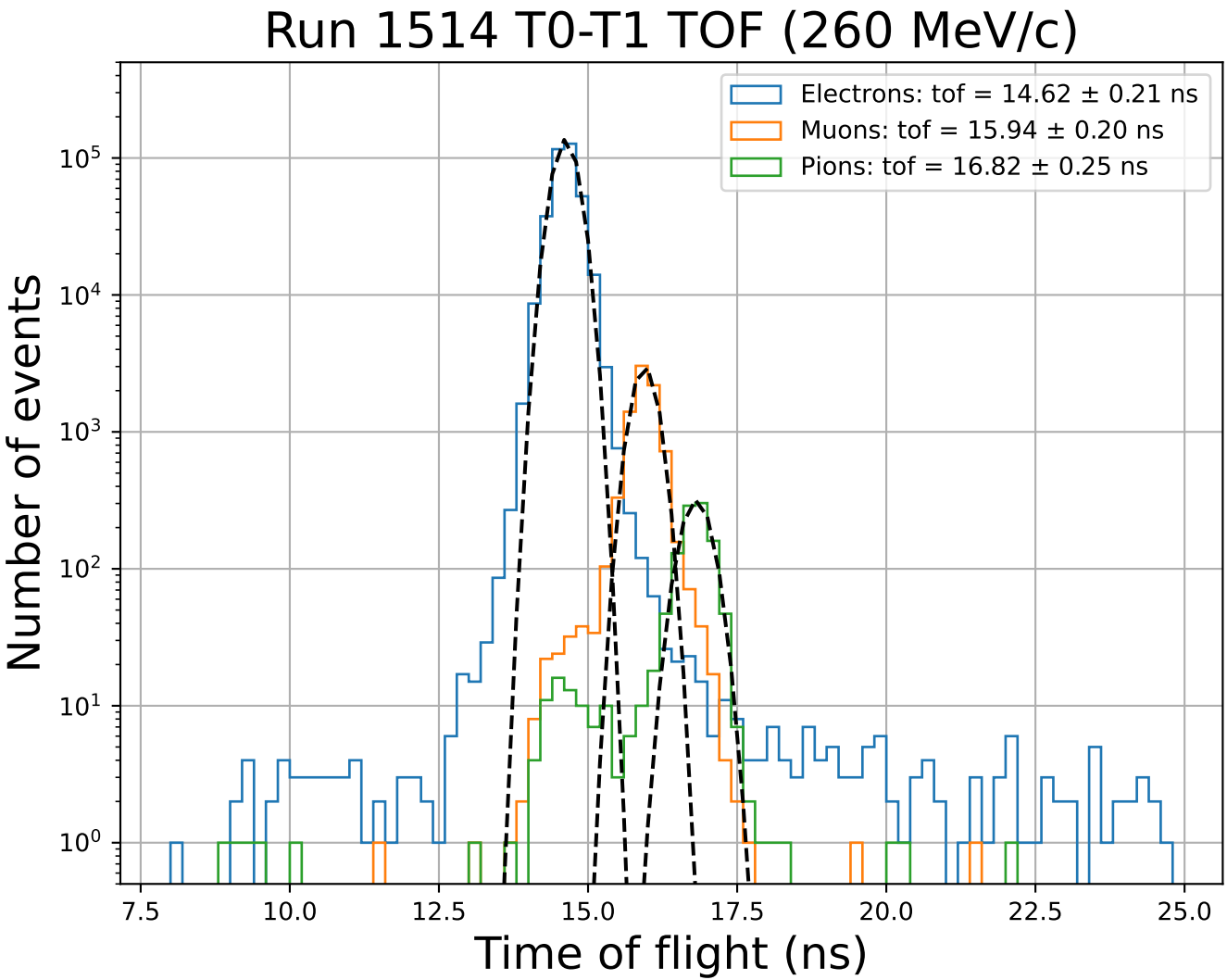

Run 1514 (260 MeV/c)
ACT3-5 without muon tagger cut
pion/muon cut line: ACT3-5 = 2.8 PE
Muons and pions
Electrons
Number of events
ACT3-5 total charge (PE)
10^0
10^1
10^2
10^3
10^4
0
10
20
30
40
50
60
70
80

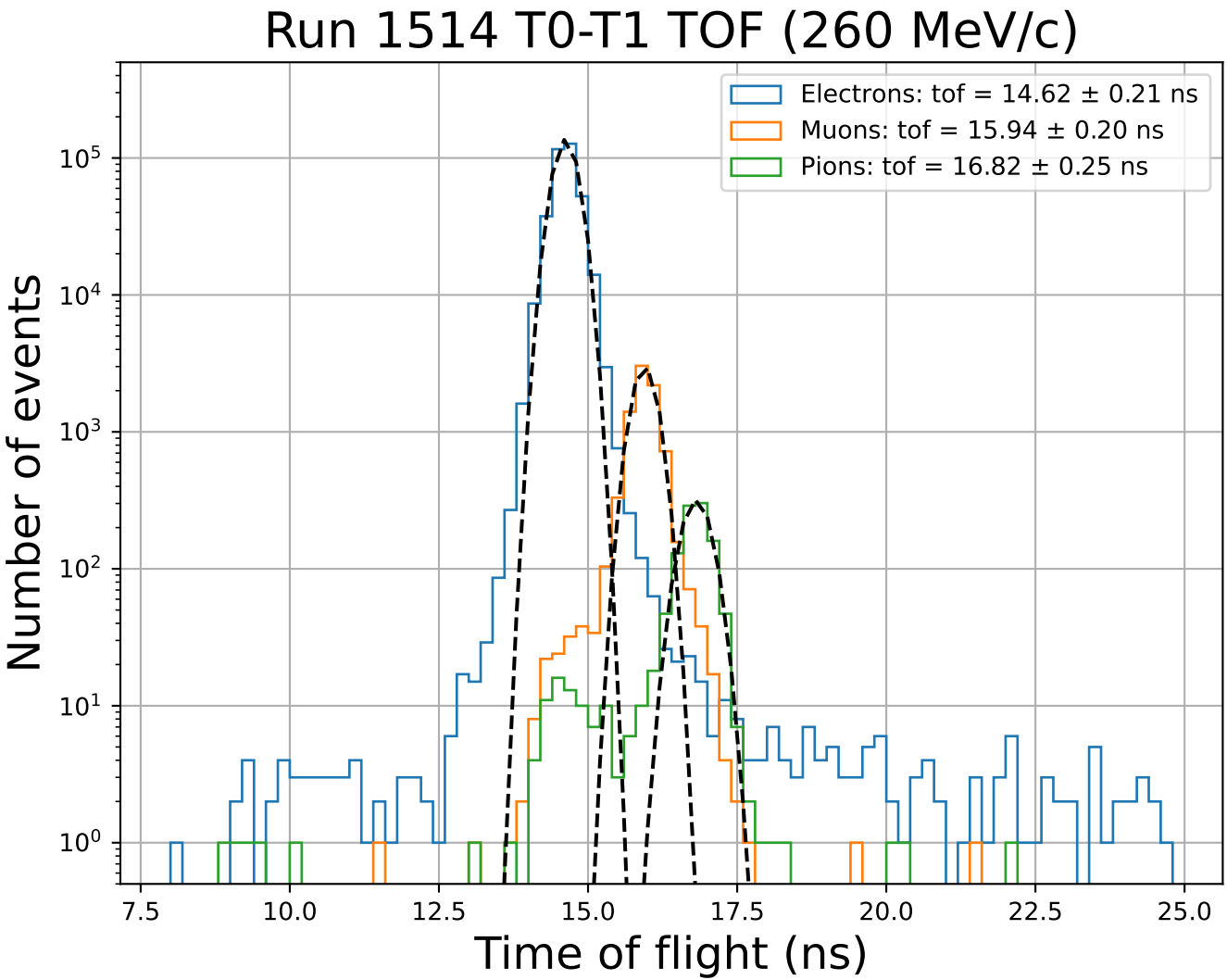

Run 1514 (260 MeV/c)
ACT3-5 all particles
pion/muon cut line: ACT3-5 = 2.8 PE
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^0
10^1
10^2
10^3

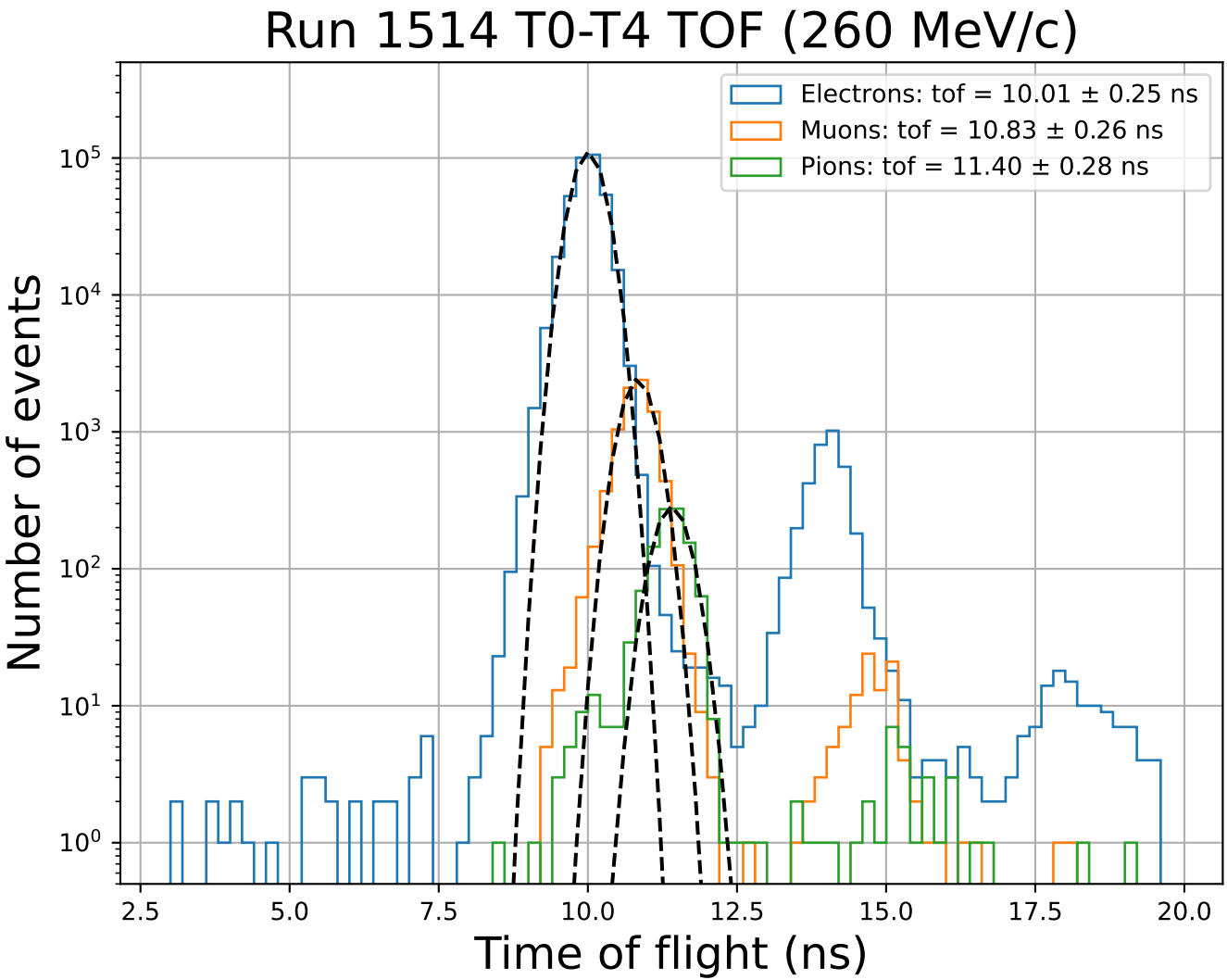

Run 1514 (260 MeV/c)
ACT35 After eveto
pion/muon cut line: ACT3-5 = 2.8 PE
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^2
10^1
10^0

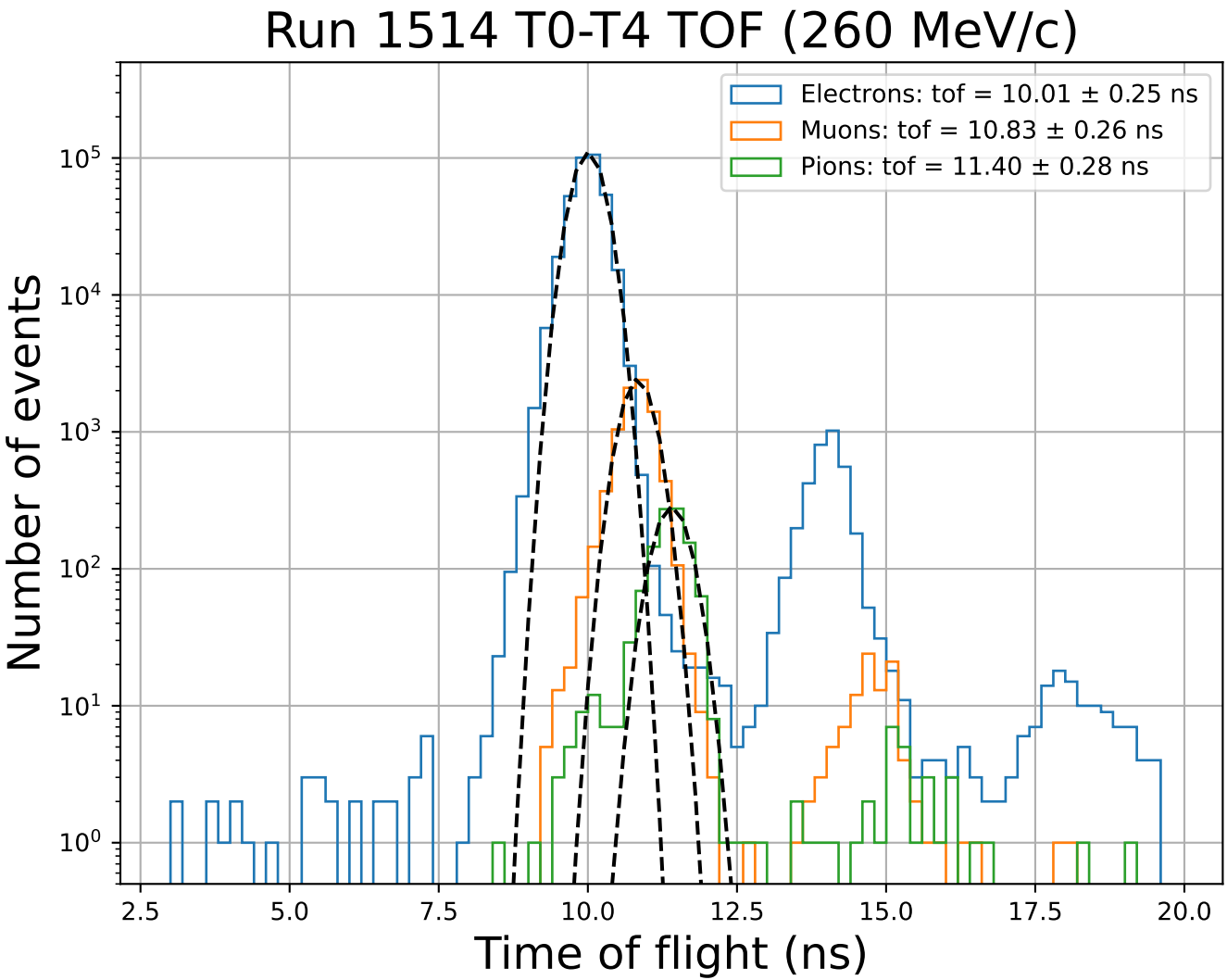

Run 1514 (260 MeV/c)
ACT3-5 After eveto and p removal
pion/muon cut line: ACT3-5 = 2.8 PE
ACT3-5 right (PE)
ACT3-5 left (PE)
0
10
20
30
40
50
60
70
10^2
10^1
10^0

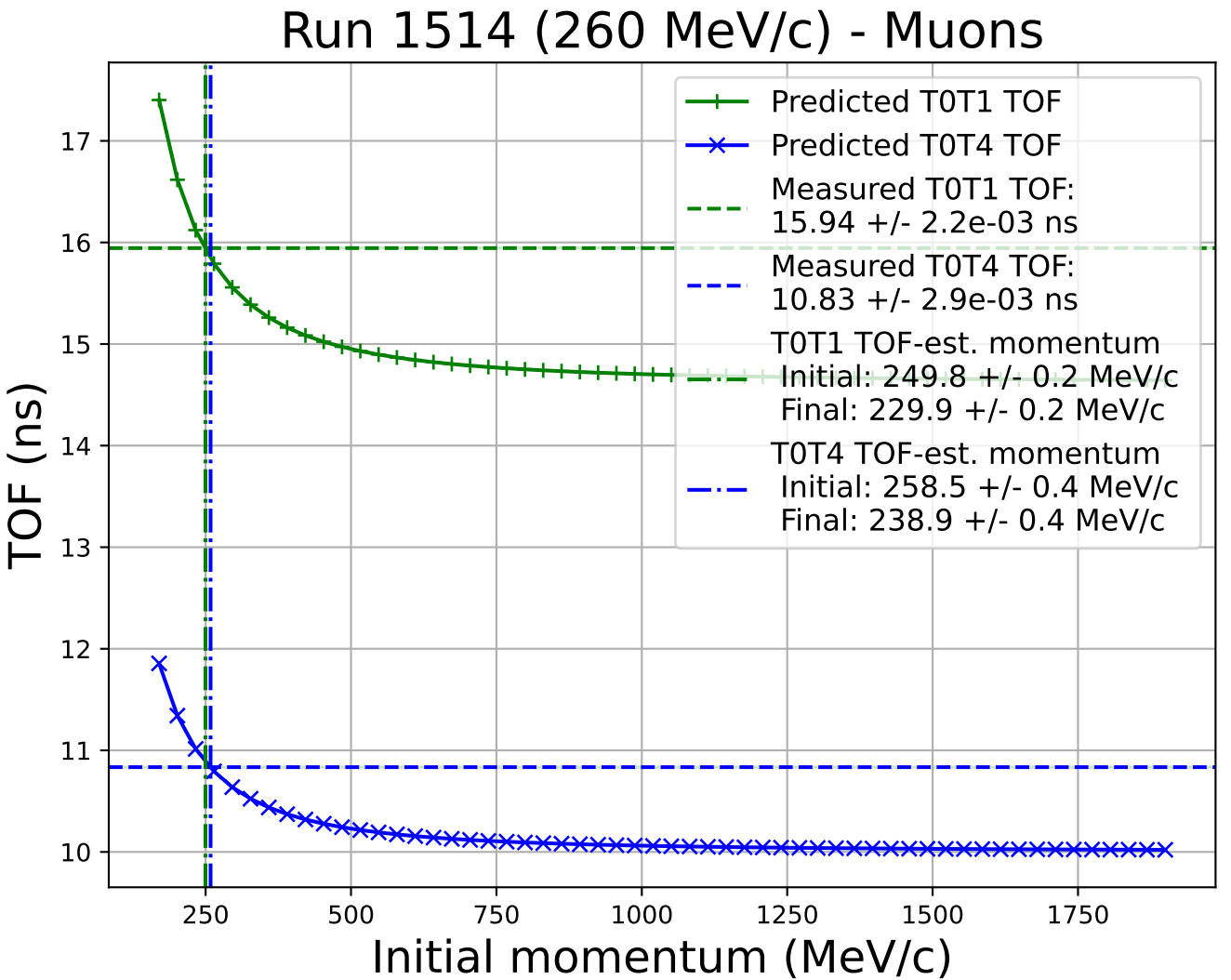

Electron T0-T4 TOF
100000
80000
60000
40000
20000
0
17.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
T0-T4 TOF (ns)

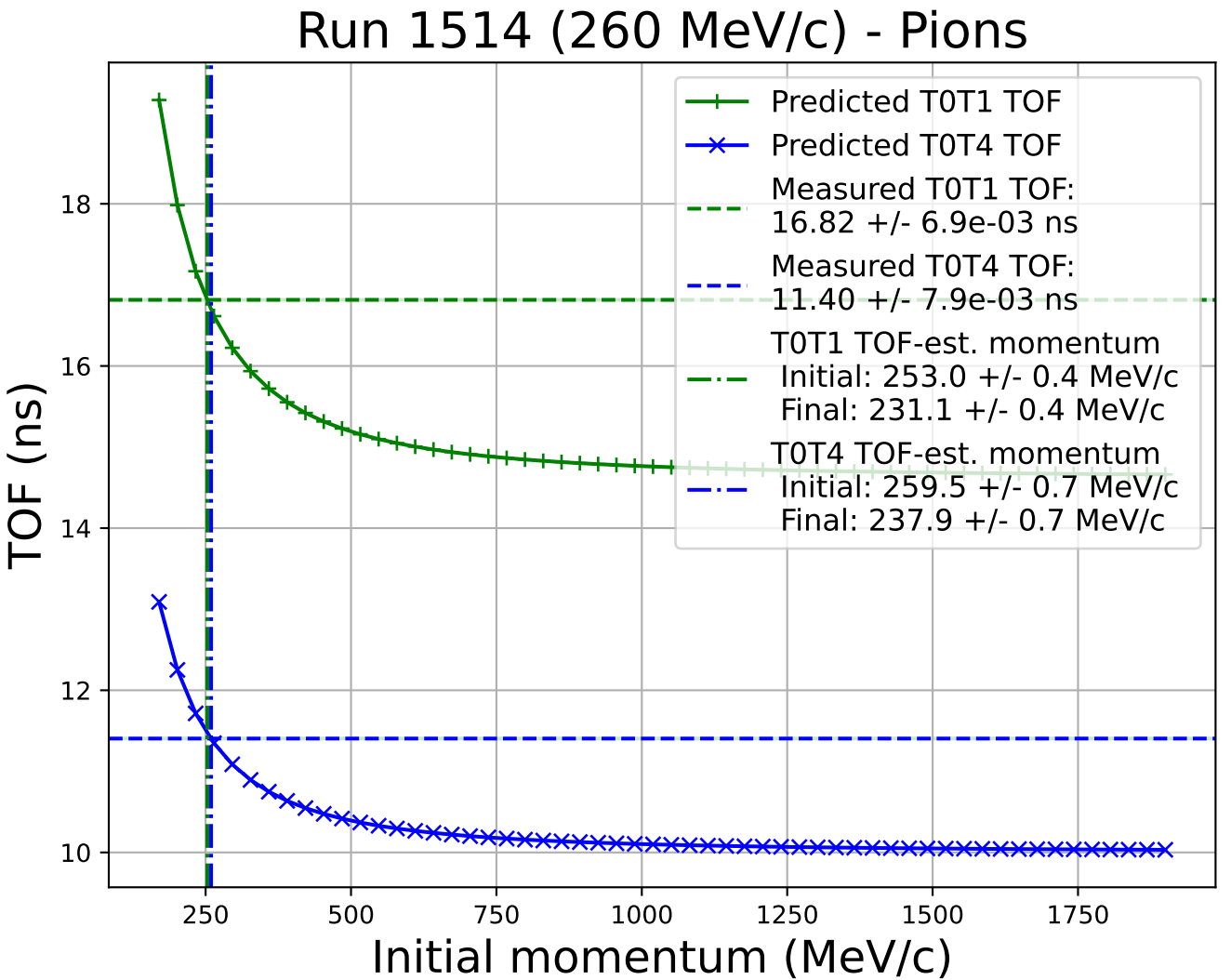

Run 1514 T0-T1 TOF (260 MeV/c)
Electrons: tof = 14.62 ± 0.21 ns
Muons: tof = 15.94 ± 0.20 ns
Pions: tof = 16.82 ± 0.25 ns
Number of events
Time of flight (ns)
10^0
10^1
10^2
10^3
10^4
10^5
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0

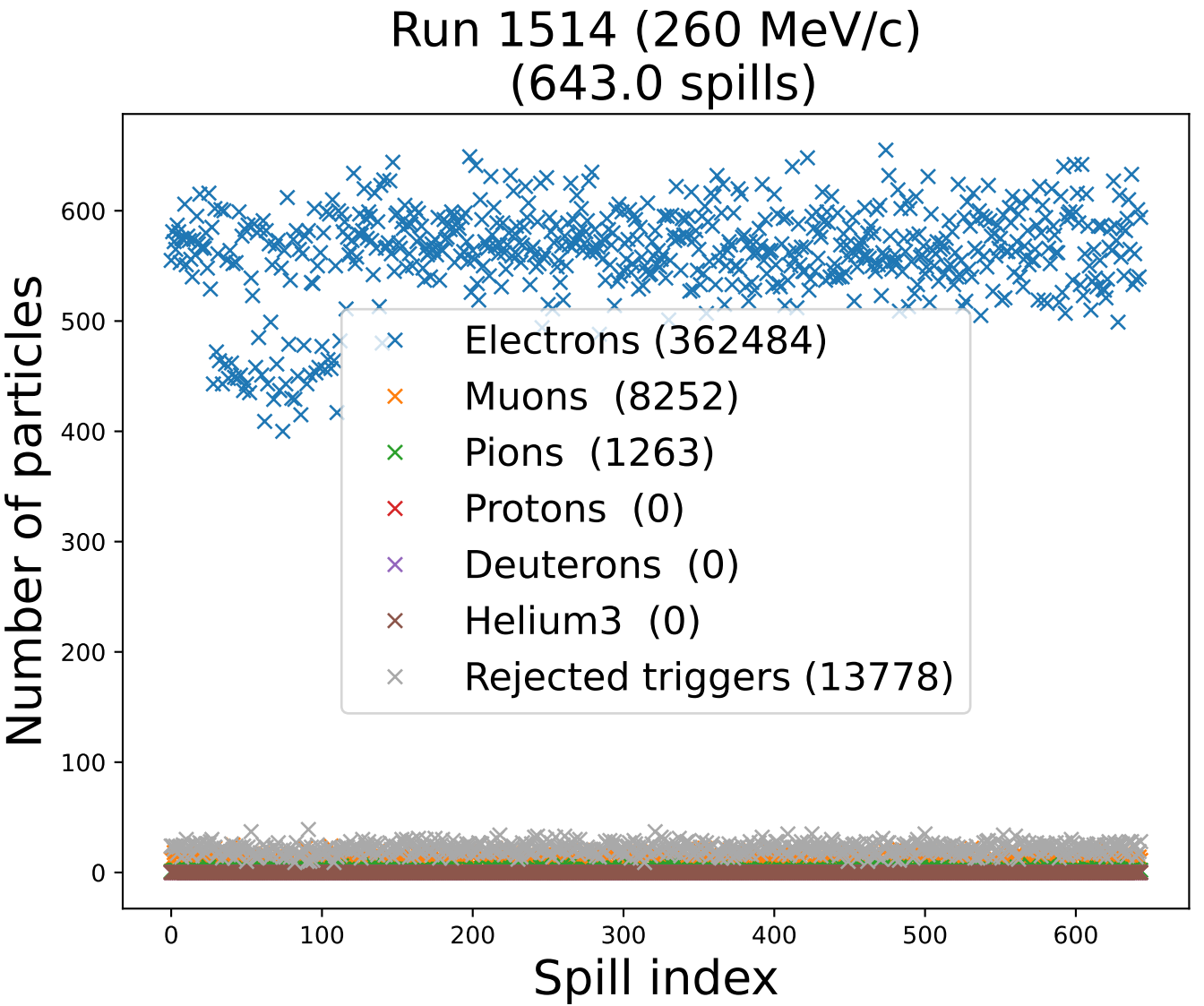

Run 1514 T0-T1 TOF (260 MeV/c)
Electrons: tof = 14.62 ± 0.21 ns
Muons: tof = 15.94 ± 0.20 ns
Pions: tof = 16.82 ± 0.25 ns
Number of events
Time of flight (ns)
10^0
10^1
10^2
10^3
10^4
10^5
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0

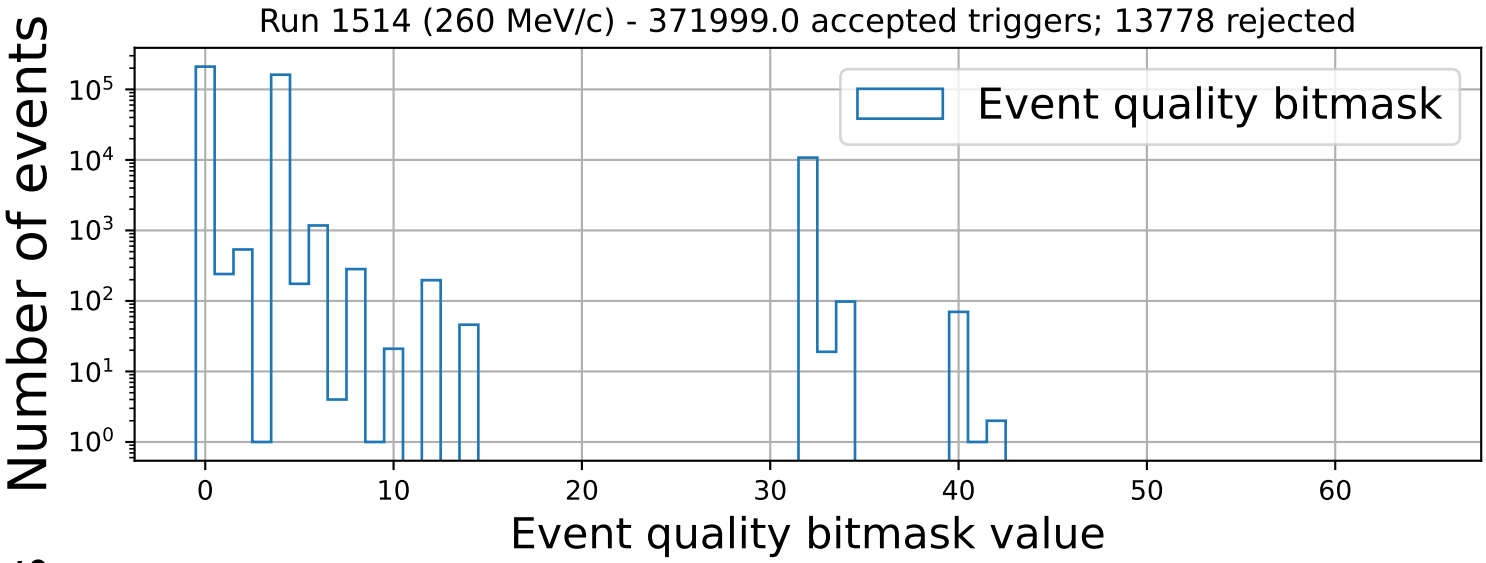

Run 1514 T0-T4 TOF (260 MeV/c)
Electrons: tof = 10.01 ± 0.25 ns
Muons: tof = 10.83 ± 0.26 ns
Pions: tof = 11.40 ± 0.28 ns
Number of events
Time of flight (ns)
10^0
10^1
10^2
10^3
10^4
10^5
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

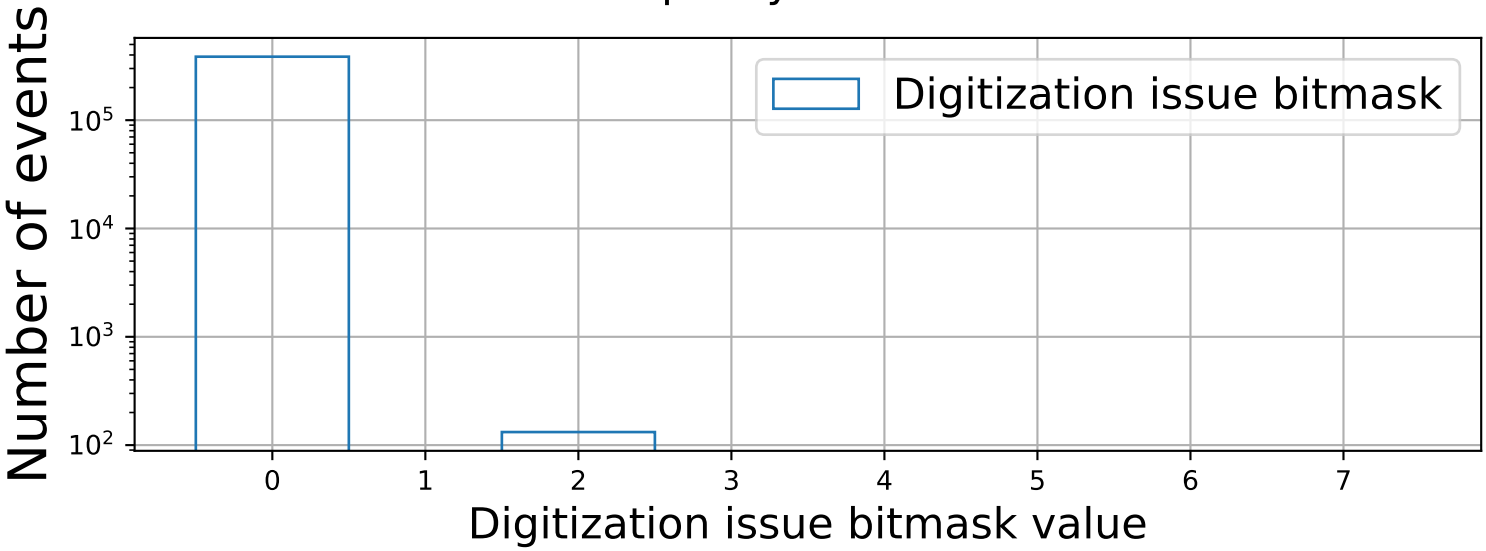

Run 1514 T0-T4 TOF (260 MeV/c)
Electrons: tof = 10.01 ± 0.25 ns
Muons: tof = 10.83 ± 0.26 ns
Pions: tof = 11.40 ± 0.28 ns
Number of events
Time of flight (ns)
10^0
10^1
10^2
10^3
10^4
10^5
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Run 1514 (260 MeV/c) - Muons
Predicted T0T1 TOF
Predicted T0T4 TOF
Measured T0T1 TOF:
15.94 +/- 2.2e-03 ns
Measured T0T4 TOF:
10.83 +/- 2.9e-03 ns
T0T1 TOF-est. momentum
Initial: 249.8 +/- 0.2 MeV/c
Final: 229.9 +/- 0.2 MeV/c
T0T4 TOF-est. momentum
Initial: 258.5 +/- 0.4 MeV/c
Final: 238.9 +/- 0.4 MeV/c
TOF (ns)
Initial momentum (MeV/c)
10
11
12
13
14
15
16
17
250
500
750
1000
1250
1500
1750


Run 1514 (260 MeV/c) - Pions
Predicted T0T1 TOF
Predicted T0T4 TOF
Measured T0T1 TOF:
16.82 +/- 6.9e-03 ns
Measured T0T4 TOF:
11.40 +/- 7.9e-03 ns
T0T1 TOF-est. momentum
Initial: 253.0 +/- 0.4 MeV/c
Final: 231.1 +/- 0.4 MeV/c
T0T4 TOF-est. momentum
Initial: 259.5 +/- 0.7 MeV/c
Final: 237.9 +/- 0.7 MeV/c
TOF (ns)
10
12
14
16
18
250
500
750
1000
1250
1500
1750
Initial momentum (MeV/c)


Run 1514 (260 MeV/c)
(643.0 spills)
Number of particles
Spill index
Electrons (362484)
Muons (8252)
Pions (1263)
Protons (0)
Deuterons (0)
Helium3 (0)
Rejected triggers (13778)
0
100
200
300
400
500
600


Run 1514 (260 MeV/c) - 371999.0 accepted triggers; 13778 rejected
Event quality bitmask
Number of events
Event quality bitmask value


Digitization issue bitmask
Number of events
Digitization issue bitmask value